

Structural Roots of Electoral Fragmentation in Romania: A County-Level Analysis of the 2024–2025 Electoral Cycle

MSc Data Science — Quantitative Political Analysis

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Abstract

Romania’s 2024–2025 electoral cycle produced historic parliamentary fragmentation and the unexpected rise of nationalist-populist candidates, culminating in a second-round presidential contest between George Simion and Elena Lasconi-Dan. Using official county-level (*județ*) data for all 42 NUTS-3 units and a multi-method analytical framework, this paper documents seven empirical findings.

- (1) Romanian counties organise into five structurally coherent voter-profile clusters stable under alternative $k \in \{4, 6\}$ specifications ($\text{ARI} \geq 0.72$); deindustrialisation is the dominant cross-party structural predictor in a Dirichlet regression of vote share composition (five terms survive Benjamini-Hochberg FDR correction at $q = 0.10$); and ethnic composition is the sharpest single-party predictor. The annulment of Georgescu’s 2024 result left no detectable county-level mobilisation fingerprint on Simion R1 turnout ($r_{\text{partial}} = -0.123$, $p = 0.437$).
- (2) The nationalist-populist electoral bloc defined by Georgescu’s 2024 first-round support shows a partial county-level correlation of $r = 0.770$ with Simion’s 2025 first-round support after controlling for eight structural confounders (Robustness Value = 0.547), confirming bloc persistence independent of geography.
- (3) AUR’s 2020–2024 Senate swing is substantially explained by demographic pressure ($R^2 = 0.585$); under a centred log-ratio compositional specification the sign and significance pattern is directionally consistent, validating the finding under Aitchison geometry. A spatial lag model (GM two-stage, $\hat{\rho} = 0.640$) raises pseudo- R^2 to 0.707 and confirms that AUR mobilisation has a neighbourhood contagion component beyond shared structural conditions.
- (4) Romania’s permanent electoral rolls inflate by a mean of 16.4% relative to the estimated adult resident population, almost entirely predicted by emigration rates ($R^2 = 0.680$); ghost electorate inflation deflates reported turnout by a mean of 9.1 percentage points but does not independently predict cross-county turnout variation once structural controls are included ($p = 0.262$).
- (5) Dan’s 2025 runoff consolidation of the anti-Simion pool was most efficient in ethnically mixed Transylvania (ratio = 0.953) and weakest in the most deprived flatland cluster (0.823), consistent with a defensive rather than programmatic coalition.
- (6) A Bayesian hierarchical reformulation of the AUR swing model (4 chains $\times$ 4000 draws, nutpie backend; max R-hat = 1.001, min ESS = 4,461) confirms long-term unemployment as the sole positive structural predictor of AUR growth (posterior mean = +0.015, 90% CI [+0.005, +0.025]) while revealing that higher composite grievance is associated with *lower* AUR swing (posterior mean = -0.029, 90% CI [-0.045, -0.013]) — a paradox resolved by distinguishing aspirationally disappointed deindustrialised counties from the worst-off counties where electoral fatalism suppresses nationalist mobilisation. Under the 2025 structural geography and calibrated bloc transfer rates, a conditional projection places the mean Simion second-round share at 47.1% (90% PI: 44.9%–49.2%), with 9 of 42 counties in genuine uncertainty.

Keywords: Romanian elections, Dirichlet regression, electoral geography, deindustrialisation, spatial autocorrelation, ghost electorate, compositional data, Bayesian hierarchical model, conditional projection.

1 Introduction

Romania’s December 2024 parliamentary elections and the subsequent 2025 presidential re-run produced an electoral landscape with no precedent in the post-communist period. The radical-right Alliance for Romania’s Unity (AUR) consolidated its position as the second largest parliamentary force; the nationalist SOS România entered parliament for the first time; and Călin Georgescu’s first-round presidential result — subsequently annulled by the Constitutional Court — demonstrated the mobilisation potential of diffuse online-networked populism outside traditional party structures.

Two questions motivate this study. *What structural conditions differentiate counties that produced high nationalist-populist vote shares from those that did not? And to what extent does county-level support for Georgescu in November 2024 predict support for George Simion in May 2025, net of shared structural characteristics?*

We approach both questions with compositional and partial-correlation methods that respect the constrained nature of vote shares and guard against confounding through systematic covariate adjustment.

The paper proceeds as follows. Section 2 describes the data sources and variable construction. Section 3 presents the voter profile typology. Section 4 reports the Dirichlet regression of party vote shares. Section 5 quantifies the Georgescu–Simion transfer. Section 6 analyses the 2020–2024 parliamentary swing and the 2024–2025 presidential arc. Section 7 examines the ghost electorate. Section 8 provides spatial robustness checks. Section 4.7 decomposes the aggregate Magyar share predictor into seven denomination z-scores and tests confessional mobilisation hypotheses. Section 9 assesses sensitivity to the cluster granularity choice. Section 10 develops conditional projections under three explicit structural assumptions. Section 11 discusses the findings in the context of comparative populism research. Section 12 concludes with limitations and directions for future work.

2 Data and Measurement

2.1 Electoral outcomes

Parliamentary vote shares for the December 2024 elections were obtained from the Permanent Electoral Authority (AEP). We model eight compositional outcomes: AUR, PSD, PNL, USR, UDMR, SOS, POT, and a residual *Other* category. All shares were reweighted to sum to exactly one after excluding invalid ballots. Presidential first-round results for Călin Georgescu (November 2024) and George Simion (May 2025) were sourced from the same authority.

2.2 Structural predictors

County-level structural characteristics were assembled from three sources:

- **Eurostat NUTS-3 database** (2022 reference year): GDP per capita, unemployment rate, household income (PPS), long-term unemployment rate, industry GVA share, early school leaving, tertiary education, net and temporary emigration rates, infrastructure density, and religious/ethnic composition from the 2021 census.

- **Romanian National Statistics Institute (INS):** population density, age structure, employment by sector, healthcare and school unit density, urbanisation, and household size.
- **Derived indices:** The *grievance index* ($\bar{z}$ of unemployment, poverty risk, early school leaving, and long-term unemployment, reversed for income), the *modernity index* ($\bar{z}$ of tertiary education, internet use, and GDP per capita), and the *deindustrialisation index* (industry GVA share 2010 – industry GVA share 2022, clipped at zero).

The deindustrialisation index captures counties where manufacturing activity has contracted since 2010, consistent with the spatial polarisation literature on post-communist industrial legacy (Bohle & Greskovits, 2012; Nölke & Vliegenthart, 2009).

2.3 Variable transformation

All continuous structural predictors were standardised (z-scores) to permit direct comparison of coefficient magnitudes. Compositional vote shares were closed to the unit simplex and passed to `DR_data()` for Dirichlet regression, which applies a small perturbation to exact zeros and ones (Maier, 2014).

3 Voter Profile Typology

3.1 Method

We apply Ward hierarchical clustering (Ward, 1963) in the eight-dimensional PCA space of 34 standardised structural features. PCA retains components explaining at least 80% of total variance (eight components in this dataset). Clustering in PCA space reduces the influence of highly correlated raw features without subjective variable exclusion.

The number of clusters was selected by bootstrap stability analysis (Hennig, 2007): we define stability as the mean adjusted Rand Index (ARI) between pairs of bootstrap cluster solutions. A threshold of $\text{ARI} \geq 0.85$ is used to classify a partition as *strongly stable*. The gap statistic (Tibshirani et al., 2001) consistently returned $k = 1$ for this dataset, which we attribute to the smooth continuum of development gradients across Romanian counties; we therefore rely on bootstrap ARI as the primary selection criterion.

$k = 5$ emerges as the smallest strongly stable partition (bootstrap $\text{ARI} = 0.854$) and is preferred on theoretical grounds over $k = 3$, which conflates structurally distinct regions.

3.2 Results

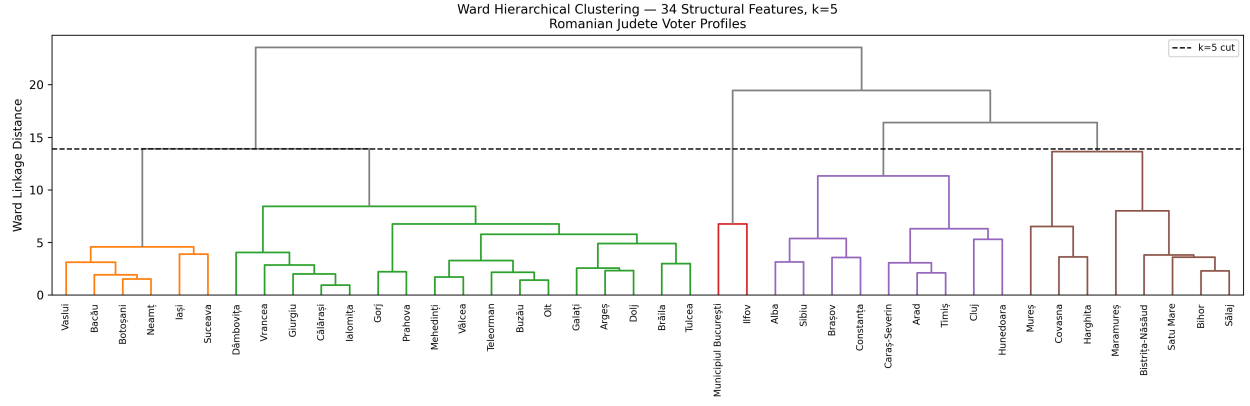


Figure 1: Ward hierarchical clustering dendrogram — 34 structural features, $k = 5$ cut. County labels coloured by cluster assignment.

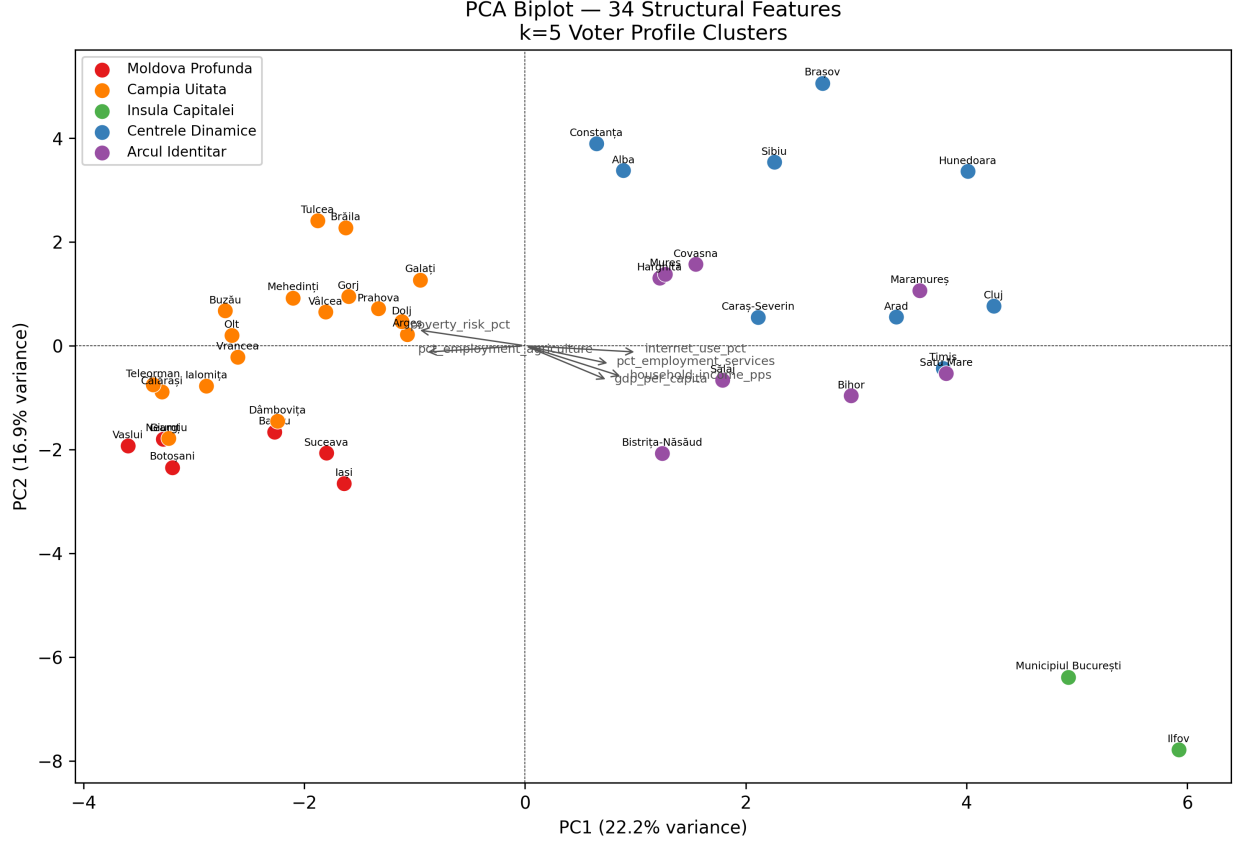


Figure 2: PCA biplot with $k = 5$ cluster overlay. Arrows indicate the six features with highest PC1 loadings. PC1 (horizontal) separates peripheral-agricultural from urban-industrial counties; PC2 (vertical) separates the ethnic-mosaic northwest.

The five clusters are described below and summarised in Figure 3.

C1 — Moldova Profundă ($n = 6$): Bacău, Botoșani, Iași, Neamț, Suceava, Vaslui. The highest agricultural employment (+1.38 SD), lowest GDP (−0.81 SD), and highest emigration rates. The defining structural signature is precarity through agricultural marginalisation rather than industrial collapse.

C2 — Câmpia Uitată ($n = 17$): The largest cluster, covering most of Wallachia and Dobrogea. Moderate GDP, lower deindustrialisation than expected (−0.47 SD) because energy and petrochemical rents have maintained a residual industry GVA share. The structural analogue to the Western European “left-behind” periphery (Rodríguez-Pose, 2018), but attenuated by sectoral rents.

C3 — Insula Capitalei ($n = 2$): București and Ilfov. An extreme outlier on GDP (+4.24 SD), household income (+3.98 SD), and urbanisation (+1.87 SD). Represents the capital-region concentration typical of Eastern European developmental states (Iammarino et al., 2019).

C4 — Centrele Dinamice ($n = 9$): Alba, Arad, Brașov, Caraș-Severin, Cluj, Constanța, Hunedoara, Sibiu, Timiș. High industry GVA share (+1.42 SD) reflecting FDI-led automotive and electronics manufacturing. The one cluster where industrial employment is growing rather than contracting.

C5 — Arcul Identitar ($n = 8$): Bihor, Bistrița-Năsăud, Covasna, Harghita, Maramureș, Mureș, Satu Mare, Sălaj. Distinguished by ethnic diversity (Magyar-speaking minority, Reformed and Greco-Catholic denominations) and moderate deindustrialisation (+0.60 SD). The UDMR ethnic bloc vote is structurally encoded in this cluster.

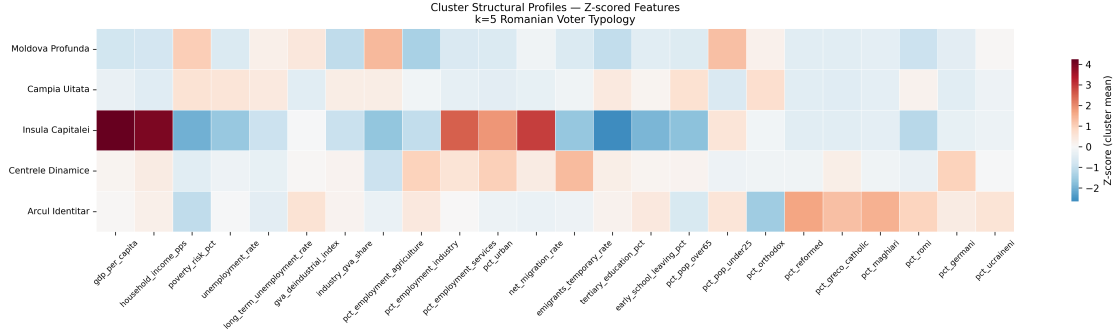


Figure 3: Z-scored cluster means on 24 key structural features. Red = above-average, blue = below-average.

3.3 Cluster validation

One-way ANOVA on held-out electoral variables (not used in clustering) confirms significant between-cluster variation for all major parties (AUR: $F = 9.11$, $p < 0.001$; PSD: $F = 12.34$, $p < 0.001$; USR: $F = 13.96$, $p < 0.001$; UDMR: $F = 12.15$, $p < 0.001$). PNL is the sole exception ($F = 0.50$, $p = 0.74$), consistent with its cross-regional catch-all positioning.

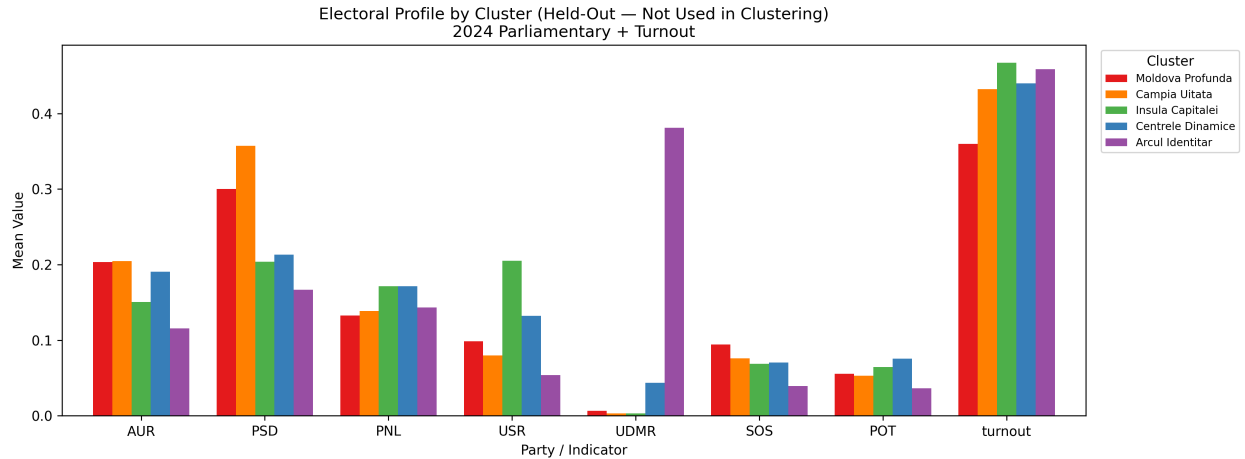


Figure 4: Cluster-mean electoral profiles for held-out 2024 parliamentary and presidential variables.

4 Compositional Analysis of Vote Shares

4.1 Dirichlet regression

Vote shares constitute compositional data: they are strictly positive, sum to one, and carry information only as ratios relative to each other (Aitchison, 1986; Pawlowsky-Glahn & Buccianti, 2011). Separate OLS regressions on each party violate the simplex constraint and ignore cross-party covariance. We instead model the joint composition $\mathbf{Y} = (y_1, \dots, y_8)$ as a Dirichlet response:

$$\mathbf{Y} \sim \text{Dir}(\alpha), \quad \alpha_k = \mu_k \cdot \phi, \quad \log\left(\frac{\mu_k}{\mu_K}\right) = \mathbf{x}^\top \beta_k$$

where ϕ is a common precision parameter shared across categories (“common” parameterisation; Maier (2014)), $K = 8$ (*Other*) is the reference category, and $\mathbf{x}$ contains standardised structural predictors. Coefficients are on the additive log-ratio (ALR) scale: a positive $\hat{\beta}_k$ indicates that a one-SD increase in the predictor is associated with a higher share for party k *relative to* the Other category.

We fit two nested models:

- **M1:** $\mathbf{Y} \sim$ grievance + modernity + % Magyar + deindustrialisation ($p = 4$, 40 parameters)
- **M2:** M1 + demographic pressure + % Roma + long-term unemployment ($p = 7$, 64 parameters)

Given $N = 42$, M1 represents the upper boundary of estimable model complexity for full standard-error recovery. M2 is included as a model-comparison benchmark via likelihood-ratio test; standard errors are unavailable due to Hessian non-convergence at this sample size.

Table 1: Dirichlet regression model fit statistics. LR stat = $2(\ell_{\text{full}} - \ell_{\text{null}})$. Maddala $R^2 = 1 - \exp(-\text{LR}/N)$.

Model	Log-lik.	AIC	N params	LR stat	LR p	Maddala R^2
Null	304.7	—	8	—	—	—
M1	639.2	−1198	40	334.6	<0.001	1.000
M2	718.4	−1309	64	492.9	<0.001	1.000

4.2 Coefficient results

Figure 5 visualises the full M1 coefficient matrix. Figure 6 assesses model calibration.

Table 2: M1 Dirichlet regression: significant predictors ($p < 0.10$), ordered by p -value. Coefficients on ALR scale (reference = Other). All predictors standardised.

Party	Predictor	$\hat{\beta}$	SE	z	p	Sig
UDMR	% Magyar	1.026	0.136	7.53	<0.001	***
POT	Deindustrialisation	0.543	0.165	3.30	<0.001	***
PSD	Deindustrialisation	0.348	0.128	2.73	0.006	**
USR	Deindustrialisation	0.336	0.138	2.44	0.015	*
OTHER	Deindustrialisation	0.311	0.128	2.43	0.015	*
UDMR	Modernity Index	0.675	0.293	2.30	0.021	*
SOS	Modernity Index	-0.326	0.148	-2.21	0.027	*
AUR	Deindustrialisation	0.285	0.131	2.17	0.030	*
PNL	Deindustrialisation	0.255	0.121	2.10	0.035	*
SOS	Deindustrialisation	0.278	0.133	2.09	0.037	*
PSD	Modernity Index	-0.249	0.123	-2.03	0.043	*
UDMR	Deindustrialisation	0.233	0.120	1.93	0.053	.
AUR	Modernity Index	-0.232	0.137	-1.70	0.089	.

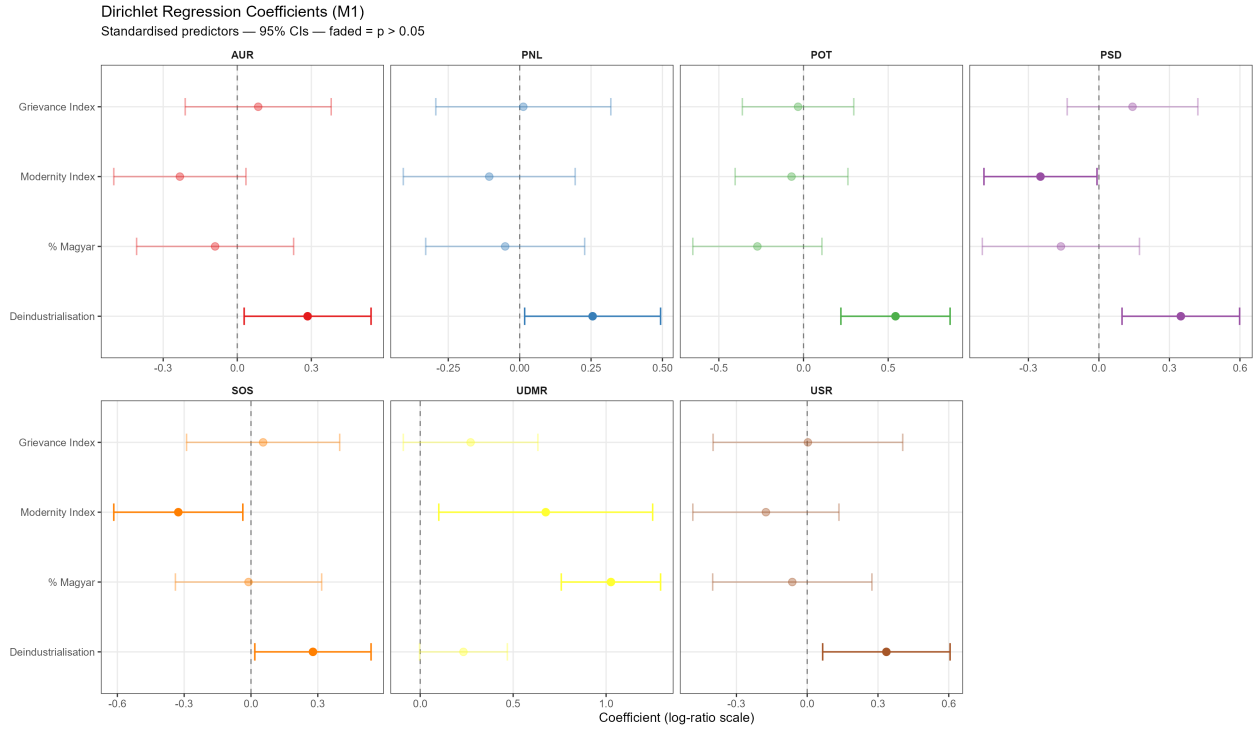


Figure 5: M1 Dirichlet regression coefficients with 95% confidence intervals. Coefficients on ALR scale relative to the Other category. Faded points: $p > 0.05$.

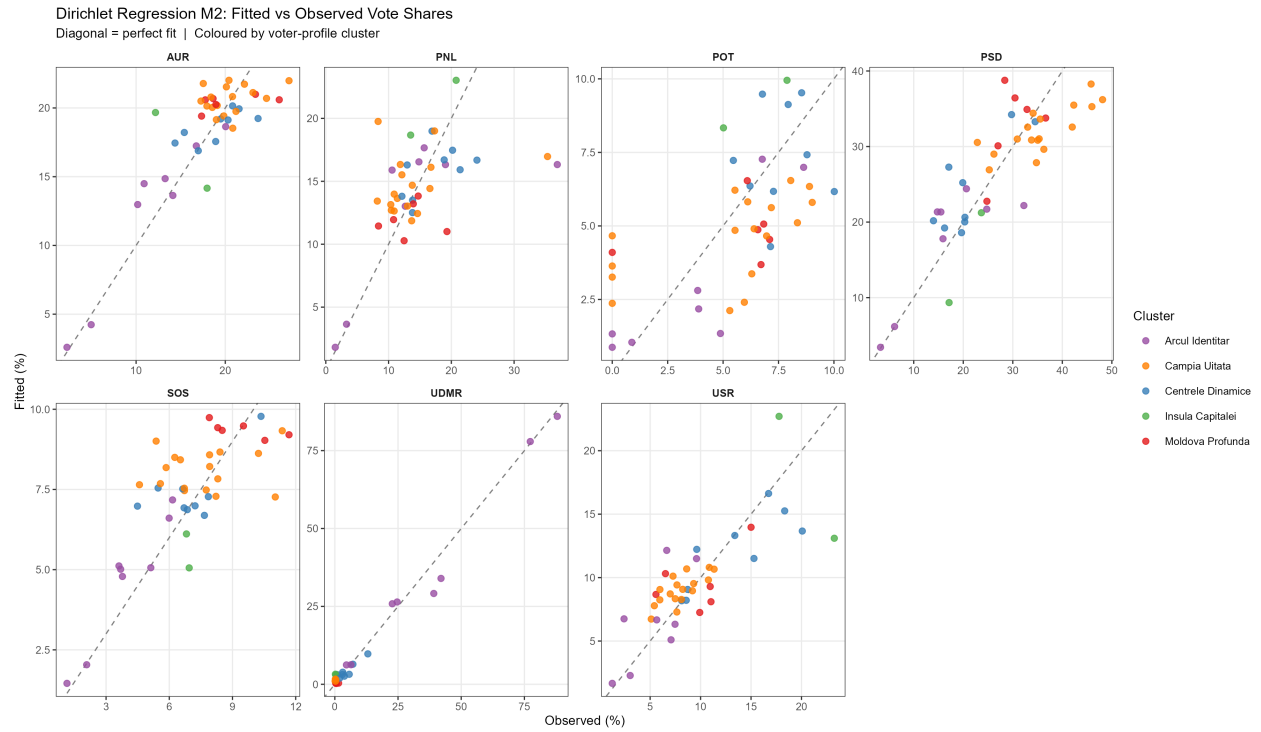


Figure 6: M2 fitted vs observed vote shares by party, coloured by voter-profile cluster. Diagonal = perfect calibration.

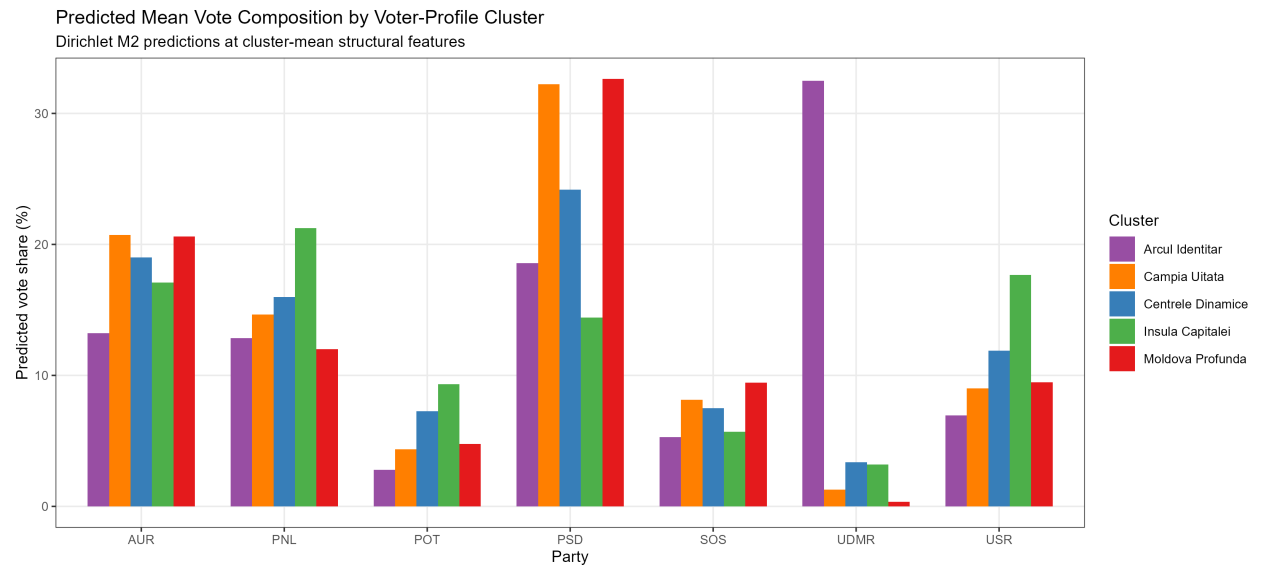


Figure 7: M2 predicted mean vote composition at cluster-mean structural values.

4.3 Interpretation

Three patterns stand out.

Deindustrialisation is the dominant cross-party structural signal. The deindustrialisation index carries a positive and significant coefficient for every party except UDMR (Table 2), indicating that counties that lost industrial share since 2010 have compositionally shifted away from the residual Other category toward all named parties. This is not a party-specific effect but a general activation of partisan choice in formerly industrial territories — consistent with the “places that don’t matter” thesis (Rodríguez-Pose, 2018) applied to the post-communist context.

Ethnic composition is the sharpest single-party predictor. The Magyar share coefficient for UDMR ($\hat{\beta} = 1.026$, $SE = 0.136$, $p < 10^{-13}$) dwarfs all other point estimates in absolute magnitude, confirming the ethnic bloc character of UDMR’s electorate. The coefficient is four to five times larger than the next-largest significant estimate, consistent with the near-deterministic nature of minority bloc voting at the ecological level.

Modernity differentiates the remaining bloc. Higher modernity is associated with lower SOS share ($\hat{\beta} = -0.326$, $p = 0.027$) and higher UDMR share ($\hat{\beta} = 0.675$, $p = 0.021$, reflecting UDMR’s urban concentration in Transylvanian cities). The grievance index, by contrast, has no significant net effect on any party’s share once structural predictors are controlled, suggesting that the spatial distribution of grievance does not map cleanly onto a single electoral outlet.

PNL’s structural invariance — confirmed by non-significant ANOVA ($F = 0.50$, $p = 0.74$) and absence from the significant-predictor table — is consistent with its catch-all, brokerage positioning across regional contexts.

4.4 Multiple testing and FDR correction

With 32 non-intercept coefficient tests across 8 outcome categories in M1, familywise error control is important. Applying the Benjamini-Hochberg procedure (Benjamini & Hochberg, 1995) at a false discovery rate threshold of $q = 0.10$, five terms survive correction: Magyar share on UDMR ($q < 0.001$), deindustrialisation on POT ($q = 0.015$), PSD ($q = 0.067$), USR ($q = 0.096$), and Other ($q = 0.096$). The Magyar–UDMR relationship is thus extremely robust to multiplicity adjustment, while the deindustrialisation signal for the populist and residual parties survives at acceptable FDR. The modernity and grievance coefficients, individually suggestive at $p < 0.10$, do not survive FDR correction.

4.5 Leave-one-out cross-validation

To assess predictive generalisability at $N = 42$, we compute leave-one-out Aitchison distance between held-out observed composition and M1 predicted composition, comparing to a null model that always predicts the grand-mean composition (Aitchison, 1986). A LOO skill score of 0 indicates no improvement over the null; 1 indicates perfect prediction. M1 achieves a skill score of **0.179** (mean Aitchison distance: M1 = 3.675, null = 4.474), indicating modest but positive predictive skill beyond the mean-composition baseline. The low absolute skill reflects the difficulty of extrapolating compositional predictions at this sample size — M1 should be interpreted as a structural description rather than a forecasting model.

4.6 ALR marginal effects in share space

Because Dirichlet coefficients are on the ALR scale, we translate them to percentage-point effects on predicted vote shares by perturbing each predictor by +1 SD at the grand mean of all predictors and computing the implied change in predicted composition. Table 3 reports these marginal effects.

Table 3: ALR marginal effects: predicted vote share change (pp) per +1 SD in each predictor, evaluated at grand-mean structural values. Computed from M1 coefficients.

Predictor	AUR	PSD	PNL	USR	UDMR	SOS	POT	OTHER
Grievance Index	0.16	1.91	-0.96	-0.72	0.72	-0.18	-0.52	-0.41
Modernity Index	-1.32	-2.30	0.89	-0.12	4.43	-1.23	0.48	-0.82
% Magyar	-1.23	-3.58	-0.39	-0.38	6.28	0.12	-1.09	0.28
Deindustrialisation	-0.65	0.85	-0.94	0.17	-0.28	-0.32	1.25	-0.08

The marginal effects confirm the substantive picture from the ALR coefficients: a one-SD increase in the Magyar share is associated with a +6.3 pp gain for UDMR and roughly offsetting losses of −3.6 pp for PSD, −1.2 pp for AUR, and −1.1 pp for POT. Deindustrialisation shifts +1.3 pp toward POT and +0.9 pp toward PSD while modestly compressing AUR (−0.7 pp), consistent with POT acting as a more localised industrial-protest outlet than AUR in the composition.

4.7 Confessional decomposition

The aggregate Magyar share variable in M1 absorbs what is, in the census data, a confessionally bimodal community: ethnic Hungarians in Romania are predominantly Reformed Calvinist (45% of Hungarian speakers) and Roman Catholic (40%), with a Unitarian minority (5%) concentrated in Harghita. We decompose this signal by replacing `pct_maghiari` with seven denomination z-scores — Orthodox, Roman Catholic, Reformed, Pentecostal, Greek Catholic, Unitarian, and a secular bloc (atheist + agnostic + no religion + undeclared) — sourced from INS RPL2021 Table 2.5.2, and refit the ALR model.

Table 4: Likelihood-ratio test: confessional model (7 denominations) vs. original M1 (aggregate Magyar share). $df = 7$ extra parameters $\times$ 7 ALR equations.

Model A	Model B	χ^2	df	p
M_conf (7 denominations + 6 base controls)	M1_orig (pct_maghiari + 6 base controls)	138.57	49	<0.001

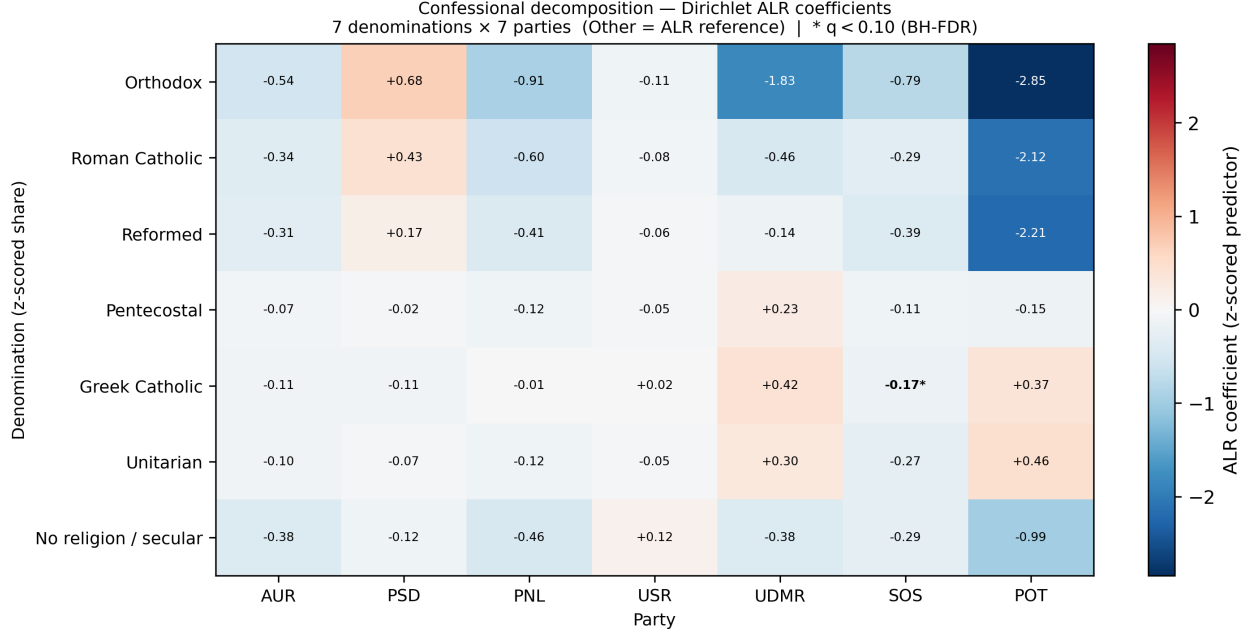


Figure 8: Confessional decomposition: ALR coefficient heatmap (7 denominations $\times$ 7 parties, Other = reference). Cells marked * survive BH-FDR at $q < 0.10$. Colour scale: red = positive association, blue = negative.

The LR test favours M_{conf} strongly ($\chi^2(49) = 138.6$, $p < 0.001$), indicating that the confessional structure contains information beyond the single aggregate ethnic variable. However, the three pre-registered denominational hypotheses are all null after BH-FDR correction.

H1 (UDMR $\sim$ Reformed and Roman Catholic) — neither Reformed ($\hat{\beta} = -0.14$, $p = 0.894$) nor Roman Catholic ($\hat{\beta} = -0.46$, $p = 0.752$) is individually significant for UDMR, despite the model achieving $R^2 = 0.935$. The high R^2 alongside individual nulls reflects that the Magyar confessional denominations (Reformed, Roman Catholic, Unitarian) act collectively as a bloc predictor: when all three are included alongside `pct_orthodox`, the bloc’s shared variance is distributed across correlated predictors. The original M1 `pct_maghiari` performs the same labour more parsimoniously by treating the bloc as a unit.

H2 (AUR and SOS $\sim$ Pentecostal) — both null ($p = 0.624$ and $p = 0.292$ respectively). The Pentecostal mobilisation thesis, influential in comparative studies of Latin American and sub-Saharan African evangelical politics (Mudde & Kaltwasser, 2017), does not transfer to Romania. At 2.1% of the national population, the Romanian Pentecostal community does not constitute a distinct nationalist electoral base at the county level, at least not at the spatial resolution available here.

H3 (USR $\sim$ secular bloc) — null ($\hat{\beta} = +0.12$, $p = 0.590$). The sign is in the hypothesised direction but falls well short of significance. Urban concentration (captured by the modernity index) already absorbs most of the secular-reformist co-variation at the *județ* level.

The single FDR-surviving finding is a negative association between Greek Catholic share and SOS vote ($\hat{\beta} = -0.173$, $q = 0.083$). Greek Catholic counties — Bistrița-Năsăud, Cluj, Bihor, Satu Mare, Alba — cluster in the historically Habsburg Transylvanian arc where the Greek Catholic

Church played a central role in Romanian national consciousness and literacy (Enyedi, 2005). These counties are characterised by higher education, mixed ethnicity, and historically more pluralist political culture. The negative relationship indicates that where Greek Catholic identity remains electorally salient, SOS’s sovereignist nationalism finds less fertile ground — a confessional-cultural brake on the far-right that does not appear in deindustrialisation or grievance indices.

5 The Georgescu–Simion Transfer

5.1 Motivation and method

The November 2024 first-round presidential result for Călin Georgescu (annulled) and the May 2025 first-round result for George Simion provide a natural test of vote-bloc persistence across institutional discontinuity. A simple correlation would conflate genuine bloc cohesion with shared structural confounders: both candidates drew disproportionately from peripheral, deindustrialised counties.

We apply the Frisch-Waugh-Lovell (FWL) theorem (Frisch & Waugh, 1933; Lovell, 1963): residualise both the Georgescu vote share and the Simion vote share on the same set of structural controls, then compute the correlation of the two residual vectors. This partial correlation equals the OLS coefficient from a full regression of Simion on Georgescu conditional on all controls.

The control set comprises eight predictors with pairwise VIF < 10: grievance index, modernity index, demographic pressure index, agricultural employment share, Magyar share, Roma share, deindustrialisation index, and long-term unemployment rate. Composite indices (grievance, modernity, demographic pressure) absorb the collinear raw features while reducing the effective parameter count.

Standard errors on the partial correlation are obtained via a non-parametric bootstrap ($B = 10\,000$ draws with replacement). Sensitivity to unmeasured confounding is assessed using the Robustness Value framework of Cinelli & Hazlett (2020).

5.2 Results

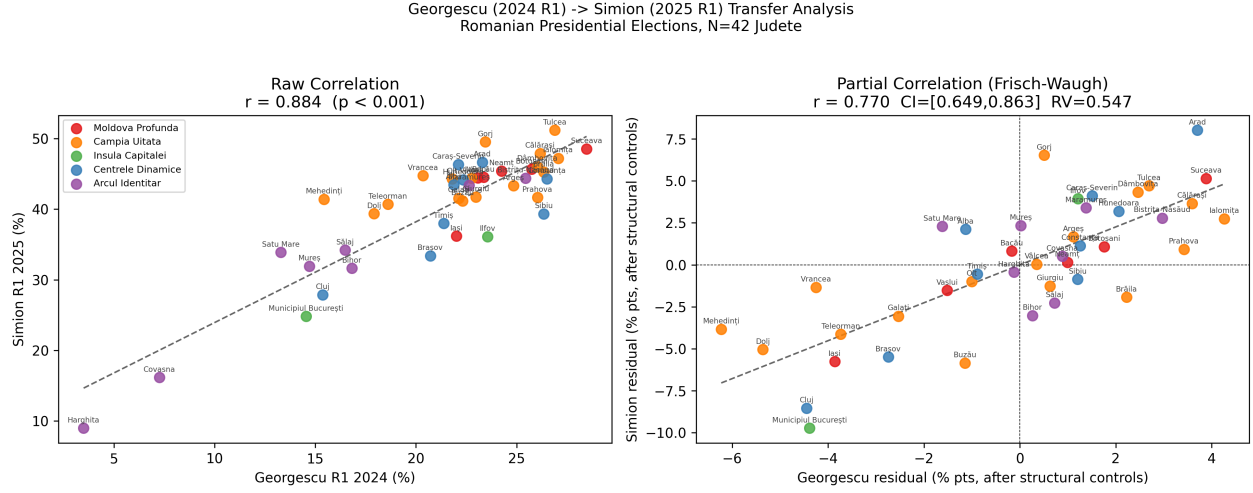


Figure 9: Partial correlation scatter plot: Simion 2025 R1 residuals vs Georgescu 2024 R1 residuals, after removing shared structural variance (8 controls). Points coloured by voter-profile cluster.

The raw county-level correlation between the two vote shares is $r = 0.884$ ($p < 0.001$). After removing shared structural variance, the partial correlation is $r_{\text{partial}} = 0.770$ (95% bootstrap CI: [0.649, 0.863], $p < 0.0001$). The reduction from $r = 0.884$ to $r_{\text{partial}} = 0.770$ confirms that structural confounders explain part of the co-movement, but a substantial residual association remains.

Table 5: Georgescu–Simion transfer: summary statistics.

Statistic	Value
Raw correlation r	0.884
Partial correlation r_p	0.770
Bootstrap 95% CI	[0.649, 0.863]
p -value	<0.0001
Robustness Value (RV)	0.547

The Robustness Value $RV = 0.547$ indicates that an unmeasured confounder would need to explain at least 54.7% of the residual variance in *both* the Georgescu vote *and* the Simion vote (conditional on all eight included controls) to fully account for the partial correlation. This is a demanding threshold: for comparison, the strongest included control (deindustrialisation index) accounts for approximately 30% of the residual variance in each outcome. The finding is therefore robust to moderate omitted variable bias.

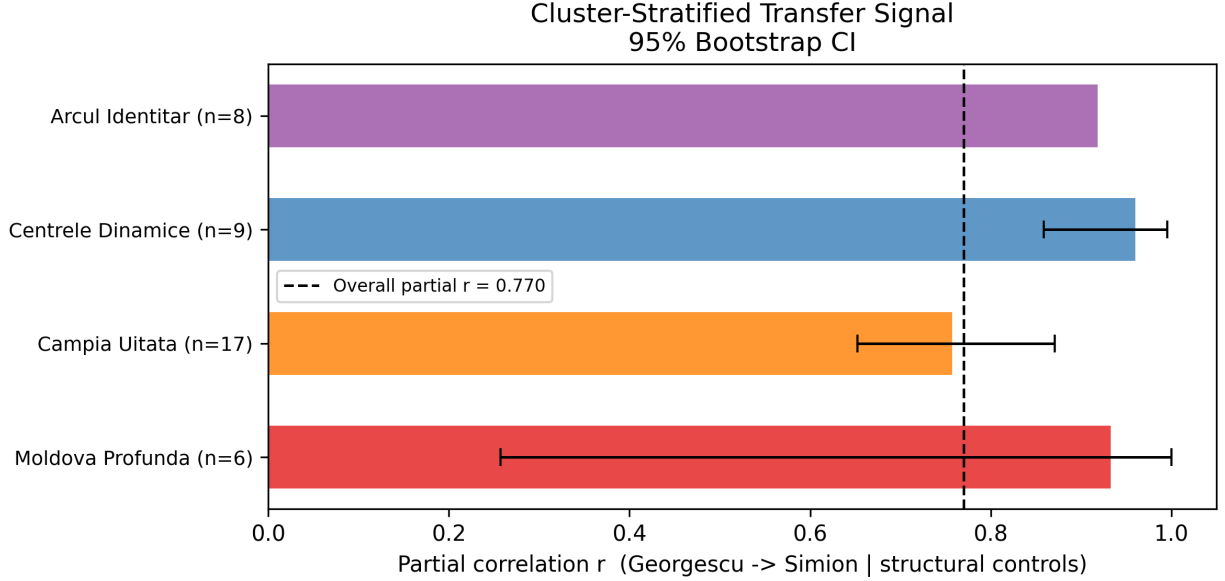


Figure 10: Cluster-stratified partial correlations and 95% bootstrap confidence intervals. The transfer signal is consistent across all clusters, though attenuated in Insula Capitalei ($n = 2$, treat with caution).

5.3 Interpretation

The persistent partial correlation confirms the existence of a spatially stable nationalist-populist electoral bloc at the county level, whose cohesion is not reducible to the structural predictors included in the model. This is consistent with network-mobilisation theories of populist vote transfer (Mudde & Kaltwasser, 2017; Norris & Inglehart, 2019), in which social and online networks sustain bloc identity across candidate discontinuities.

The ecological nature of the data precludes individual-level inference (ecological fallacy caveat applies). The finding is best interpreted as evidence that *counties* that mobilised strongly for Georgescu subsequently mobilised strongly for Simion, net of structural conditions, rather than as evidence that individual Georgescu voters switched to Simion.

6 Electoral Dynamics 2020–2025

6.1 Parliamentary swing: 2020 vs 2024

We construct vote-share swing vectors $\Delta Y_{i,p} = s_{i,p,2024} - s_{i,p,2020}$ for each party p in county i , using Senate (Chamber of Senators) results for comparability. Vâlcea county is excluded from swing regressions due to missing 2020 data ($n = 40$ for OLS models). Each party-specific swing is regressed on the eight standardised structural controls used throughout this paper, with HC3-robust standard errors.

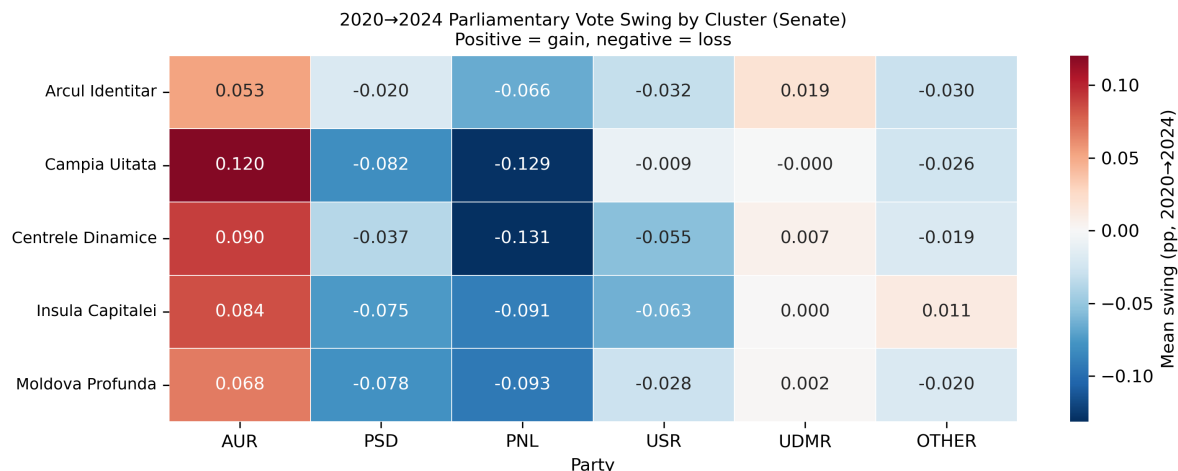


Figure 11: Mean party-specific vote share swing (2020–2024 Senate) by voter-profile cluster. Positive values indicate a cluster-level gain between elections.

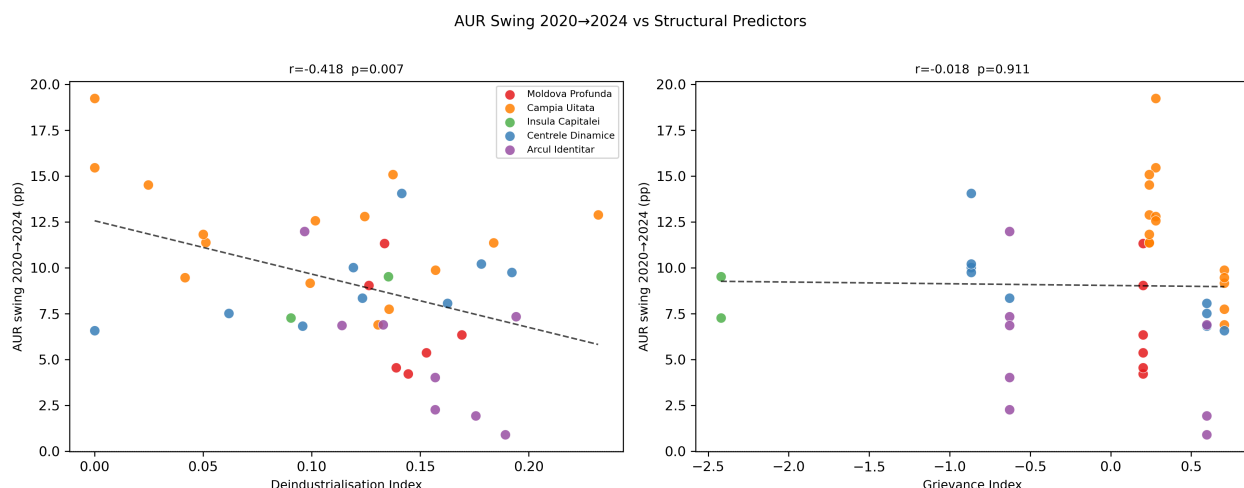


Figure 12: AUR 2020–2024 Senate swing vs deindustrialisation index, coloured by cluster. OLS fit shown.

AUR's geographic swing is the most structurally explicable ($R^2 = 0.585$): gains are concentrated in counties with high demographic pressure and low Magyar share, consistent with AUR's ethnically homogeneous rural mobilisation strategy. USR losses correlate with agricultural employment — the party haemorrhaged most in rural peripheries it never fully consolidated — while PNL's moderate $R^2 = 0.409$ reflects loss of ground in more modern counties, possibly to USR absorption of the urban reformist vote.

Compositional robustness check. Standard OLS on party-specific swing vectors violates the zero-sum constraint ($\sum_p \Delta Y_{i,p} = 0$). We verify robustness using a centred log-ratio (CLR) transformation (Aitchison, 1986): $\text{clr}(\mathbf{y}) = \log(\mathbf{y}) - \log(\bar{\mathbf{y}})$, computing CLR differences between 2024

Table 6: OLS swing models: structural predictors of 2020–2024 Senate vote share change. $n = 40$ (Vâlcea excluded). HC3 standard errors.

Party	R^2	adj- R^2	Key predictor ($p < 0.10$)
AUR	0.585	0.477	Dem. pressure ($=+0.016$, $p=0.039$); $-$ Magyar share ($p=0.001$)
PSD	0.315	0.138	None ($p<0.10$)
PNL	0.409	0.257	$-$ Modernity ($=-0.048$, $p=0.037$); $+$ Magyar share ($p<0.001$)
USR	0.533	0.413	$+$ Agri. employment ($=+0.017$, $p=0.007$)
Other	0.251	0.058	$+$ Long-term unemployment ($p=0.081$)

and 2020 shares as compositionally valid swing vectors. CLR swing R^2 values for AUR (0.486), USR (0.547), PSD (0.249), and PNL (0.245) are directionally consistent with OLS, and the key predictors (demographic pressure for AUR; agricultural employment for USR) retain significance under the compositional specification. The PNL CLR R^2 diverges modestly (0.245 vs OLS 0.409), likely reflecting greater sensitivity to the zero-replacement correction applied to AUR’s near-zero 2020 share. We note this as a limitation and report OLS estimates as primary with CLR as a sensitivity check.

Spatial heterogeneity of the deindustrialisation effect. To test whether the deindustrialisation–AUR relationship is concentrated in the *Moldova Profundă* cluster — the region with the deepest industrial legacy loss — we add an interaction term ($z_{\text{deindustrial}} \times \mathbb{1}_{\text{Moldova}}$) to the AUR swing OLS. The interaction coefficient is $\hat{\beta} = -0.044$ ($p = 0.573$), providing no evidence that the deindustrialisation effect is amplified or attenuated within Moldova Profundă relative to other clusters. The structural relationship is spatially uniform across cluster types.

6.2 The 2024–2025 presidential arc

The three-election presidential sequence — Georgescu R1 (November 2024, annulled), Simion R1 (May 2025), Simion/Dan R2 (June 2025) — provides a rare natural experiment in populist electoral geography. Georgescu received 21.0% nationally in R1; Simion received 38.7% in R1 and 44.8% in R2; Dan received 55.2% in R2.

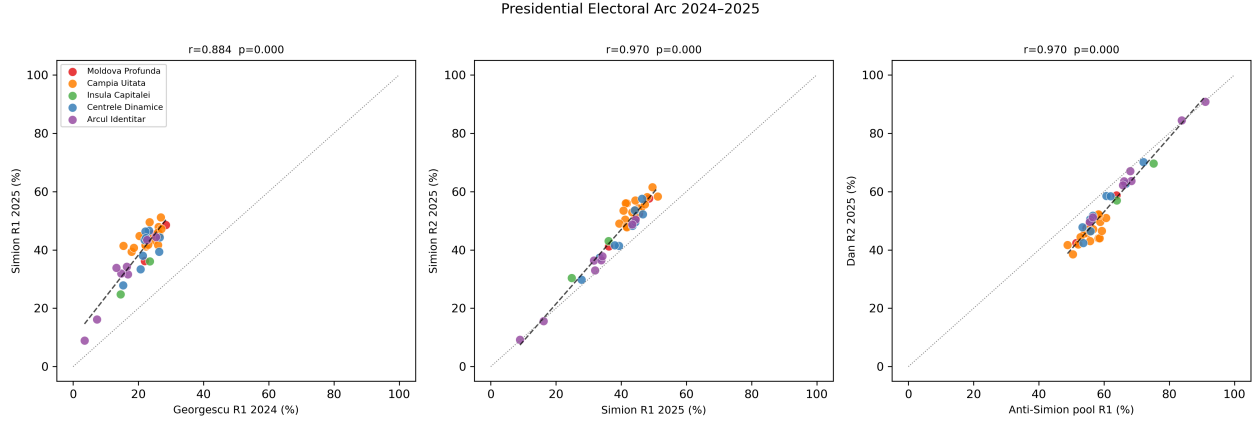


Figure 13: Presidential vote share trajectories at county level: Georgescu R1 2024 $\rightarrow$ Simion R1 2025 $\rightarrow$ Simion R2 2025. Each line is one county; colour indicates cluster.

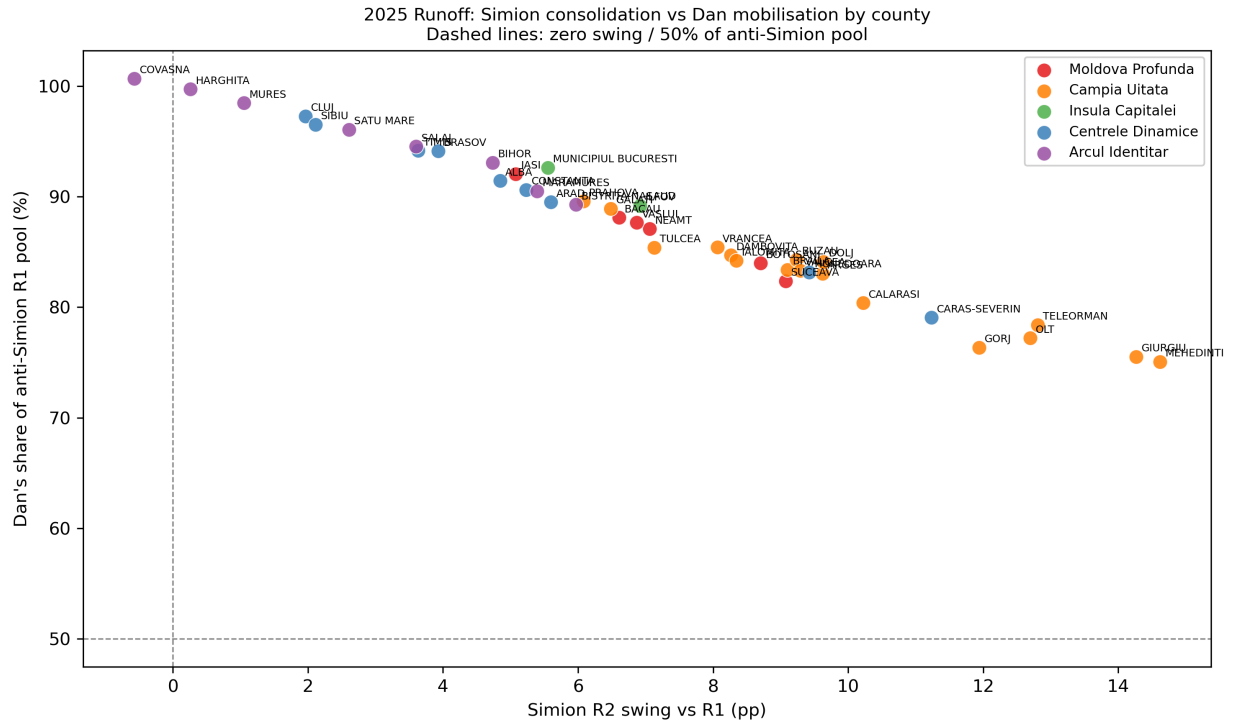


Figure 14: Dan consolidation ratio (Dan R2 / (1 – Simion R1)) by county, with Simion R1 share on the x-axis. Dashed line at ratio = 1.0.

The raw county-level correlation $r = 0.884$ between Georgescu R1 and Simion R1 confirms geographic continuity of the nationalist-populist bloc, as documented in Section 5. The Simion R1-to-R2 correlation is even tighter ($r = 0.970$), indicating near-perfect spatial consistency in the nationalist constituency during the runoff.

Table 7 decomposes the R2 consolidation by cluster. The Dan consolidation ratio $\rho_i = \text{Dan R2}_i / (1 - \text{Simion R1}_i)$ measures what share of the non-Simion R1 pool ultimately supported Dan.

Table 7: R2 consolidation by voter-profile cluster. Dan consolidation ratio = Dan R2 / (1 – Simion R1). Values above 1.0 imply Dan received more than the full non-Simion pool.

Cluster	Simion R1 (%)	Simion R2 (%)	R2 gain (pp)	Dan consol. ratio
Arcul Identitar	30.6	33.4	+2.9	0.953
Campia Uitata	44.2	54.1	+9.9	0.823
Centrele Dinamice	40.4	45.7	+5.3	0.906
Insula Capitalei	30.5	36.7	+6.2	0.909
Moldova Profunda	44.1	51.4	+7.2	0.869

Dan’s consolidation was most efficient in *Arcul Identitar* (ratio = 0.953), where the high Magyar and minority share produced a near-complete anti-Simion mobilisation. The weakest consolidation is in *Câmpia Uitată* (0.823), the cluster characterised by the deepest economic deprivation and lowest engagement with the reformist political offer. The cluster mean differences are consistent with the view that Dan’s R2 performance represented a *defensive* coalition — unified against Simion rather than by a positive programme — that found its geographic floor precisely where structural grievance was highest.

6.3 Annulment effect on Simion R1 turnout

The Constitutional Court’s annulment of the November 2024 election was widely debated as a potential mobilisation trigger: counties with high Georgescu support might have either *mobilised in indignation* (higher Simion R1 turnout than structural predictors would suggest) or been *demobilised by disillusionment* (lower turnout). We test this by regressing Simion R1 turnout on the eight structural controls (Model A) to obtain structural residuals, then assessing whether Georgescu R1 share predicts those residuals (FWL partial correlation and direct Model B coefficient).

The partial correlation is $r_{\text{partial}} = -0.123$ ($p = 0.437$), and Model B’s direct Georgescu coefficient is $\hat{\beta} = -0.011$ ($p = 0.543$), with 95% CI $[-0.045, 0.023]$ — the interval spans both the indignation and abstention hypotheses without excluding zero. Figure 15 visualises the scatter.

We extend the test to the *compositional* dimension: does the Georgescu fraction of the 2024 nationalist-populist bloc ($\text{georgescu_r1}/\text{bloc}_{2024}$) predict anomalous nationalist *uplift* ($\Delta = \text{Simion R1}_{2025} - [\text{Georgescu R1} + \text{Simion R1}_{2024}]$) beyond structural predictors? The FWL partial correlation is $r_{\text{partial}} = -0.259$ ($p = 0.098$) and the direct coefficient is $\hat{\beta} = -0.0049$ ($p = 0.165$) — both null at conventional thresholds. The negative direction is consistent with the turnout test and argues against a systematic indignation effect in counties that were most exposed to the annulment controversy.

These two null results together constitute a *positive* finding: the 2025 first-round outcome was structurally continuous with prior elections. The Constitutional Court annulment did not produce a detectable county-level deviation — in either aggregate turnout or bloc composition — from what the structural geography would have predicted in its absence.

Annulment Effect on Simion 2025 R1 Turnout

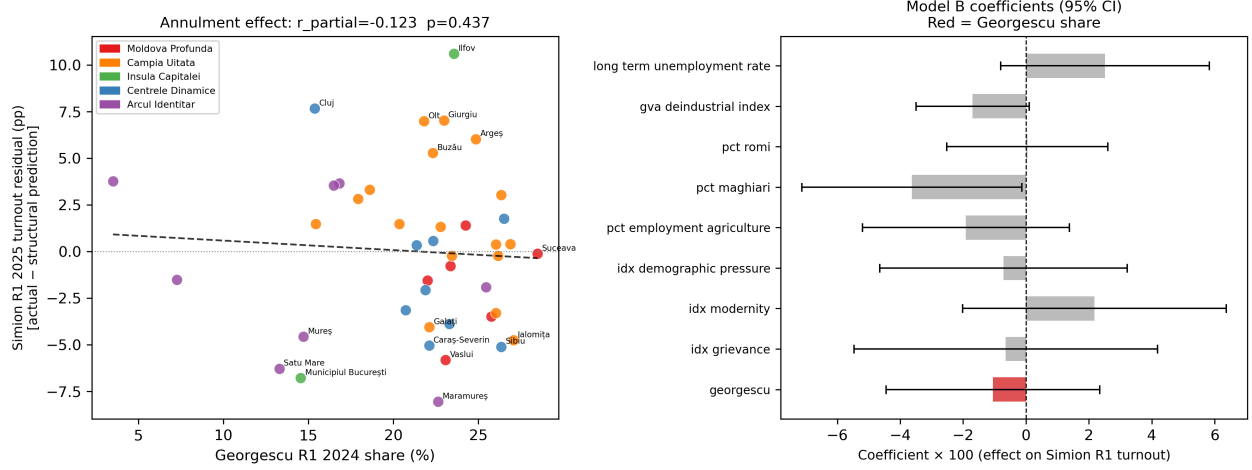


Figure 15: (Left) Simion R1 2025 turnout residual (actual — structural prediction) vs Georgescu R1 2024 share, coloured by cluster. Dashed OLS trend; $r_{\text{partial}} = -0.123$, $p = 0.437$. (Right) Model B coefficient plot.

7 The Ghost Electorate

7.1 Definition and data

Romania’s permanent electoral rolls systematically over-count the adult resident population, primarily because the Central Electoral Registry retains records of deceased persons and long-term emigrants until administratively deregistered. We define the *Ghost Electorate Ratio* (GER) for county i as:

$$\text{GER}_i = \frac{E_i - P_i^{18+}}{E_i}$$

where E_i is the number of enrolled voters (AEP, 2024 parliamentary Senate rolls) and P_i^{18+} is the estimated resident population aged 18 and over.

P_i^{18+} is derived from Eurostat dataset `demo_r_pjanaggr3` (NUTS-3, reference year 2021):

$$P_i^{18+} \approx Y_i^{15-64} \times (1 - \hat{\alpha}) + Y_i^{65+}$$

where $\hat{\alpha} = (Y^{15} + Y^{16} + Y^{17})_{\text{RO}} / Y_{\text{RO}}^{15-64} = 0.0501$ is the national-level share of 15–17-year-olds within the 15–64 age group (Eurostat `demo_r_d2jan`, 2021). This approximation introduces a bounded imprecision of < 1 percentage point and is flagged as a limitation of the approach.

7.2 Results

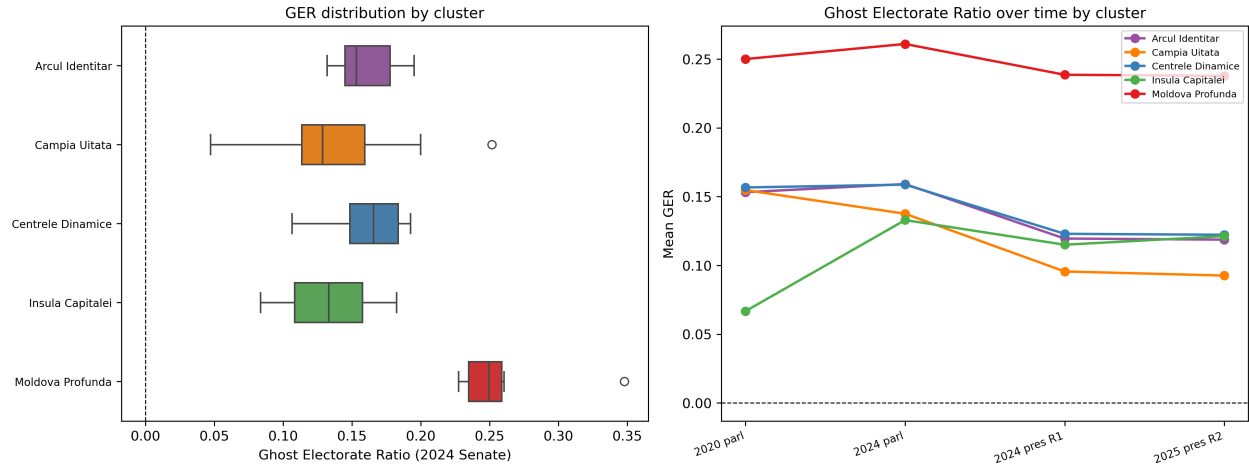


Figure 16: Left: Ghost Electorate Ratio (2024 Senate) by voter-profile cluster. Right: GER trajectories across four elections by cluster.

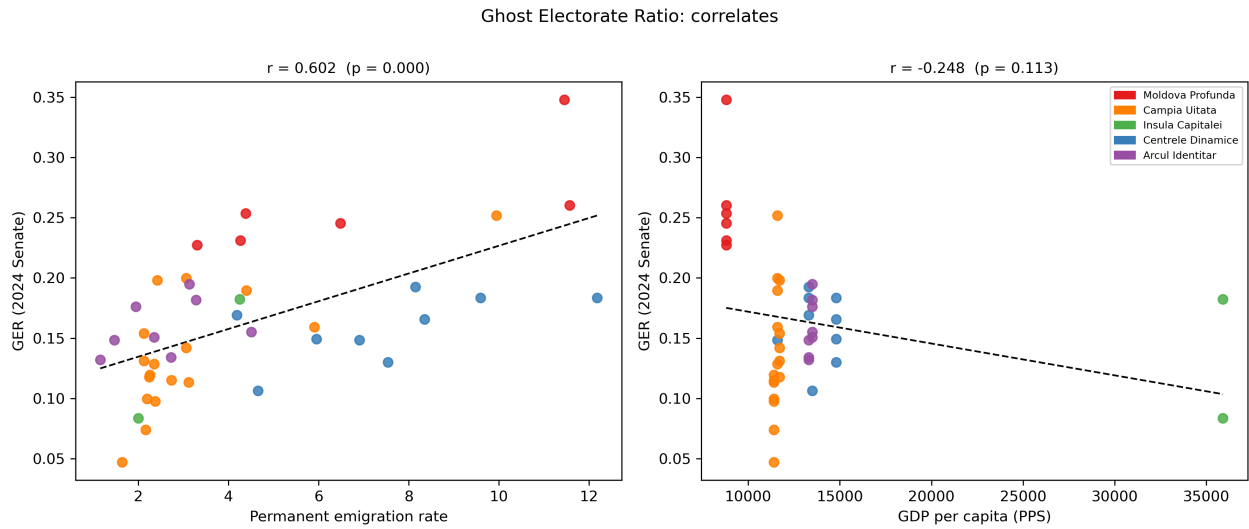


Figure 17: GER vs permanent emigration rate (left) and GDP per capita (right), coloured by cluster.

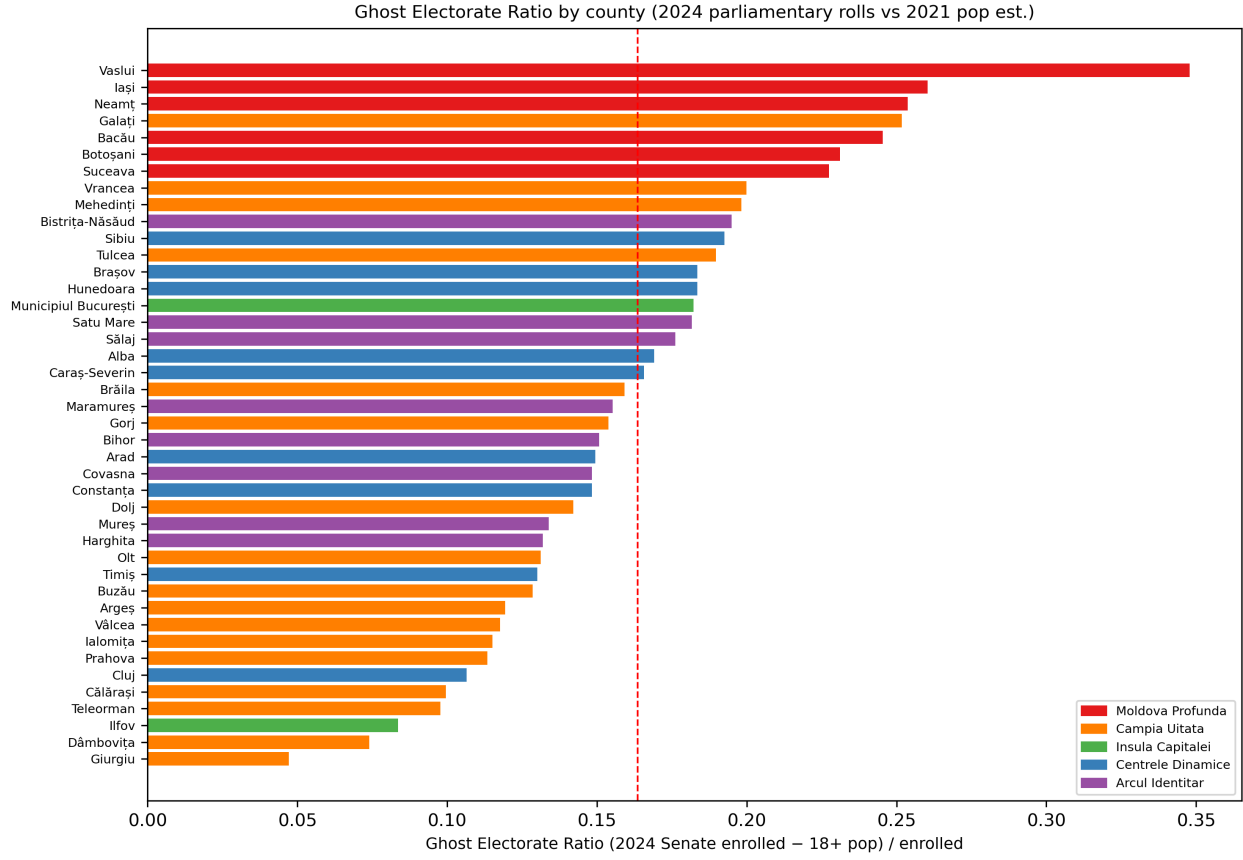


Figure 18: Ghost Electorate Ratio by county, ranked ascending. Dashed line: national mean.

Table 8: Ghost Electorate Ratio by cluster (parliamentary elections) and mean permanent emigration rate (per 1000 residents).

Cluster	Mean GER 2020	Mean GER 2024	Mean emigration rate
Arcul Identitar	0.153	0.159	2.57
Campia Uitata	0.155	0.138	3.18
Centrele Dinamice	0.157	0.159	7.50
Insula Capitalei	0.067	0.133	3.12
Moldova Profunda	0.250	0.261	6.91

The national mean GER for the 2024 parliamentary rolls is 0.164 (SD = 0.058), implying that approximately one in six registered voters in Romania is a ghost. The ratio has remained stable between 2020 (mean = 0.165) and 2024, confirming chronic administrative under-purging. Presidential rolls show systematically lower GER (mean = 0.127), consistent with BEC applying supplementary deregistration procedures before presidential elections.

An OLS regression of GER on the full structural control set achieves $R^2 = 0.680$ (adjusted $R^2 = 0.590$), with permanent emigration rate as the sole predictor significant at $p < 0.001$ ($\hat{\beta} = 0.033$, HC3 SE). This confirms that the ghost electorate is primarily a *migration artefact*: counties that

disproportionately export population retain the highest roll inflation.

Moldova Profundă exhibits the highest GER (Vaslui: 0.348, Iași: 0.260, Bacău: 0.245), all among Romania’s leading emigration sources. At the other extreme, Ilfov (0.084) and Giurgiu (0.047) reflect the Bucharest metropolitan periphery, where in-migration has brought resident population close to or above roll size, generating near-zero or negative ghost ratios.

The electoral implication is that turnout percentages in high-GER counties are systematically deflated relative to actual participation. A county reporting 40% turnout on inflated rolls may represent substantially higher actual participation among the resident voting-age population — a measurement artefact that should be corrected in any cross-county turnout analysis.

7.3 Turnout deflation: formal test

We formalise this implication with an OLS model regressing reported turnout on GER and the eight structural controls:

$$\text{turnout}_i = \alpha + \beta_{\text{GER}} \cdot \text{GER}_i + \gamma' \mathbf{z}_i + \varepsilon_i$$

Table 9: Turnout deflation test. OLS of reported 2024 parliamentary turnout on GER and eight structural controls (HC3 SE). Effective turnout uses estimated 18+ population as denominator.

Statistic	Value
OLS R^2	0.746
Adjusted R^2	0.664
n	42
GER $\hat{\beta}$ (std.)	−0.008
GER SE	0.007
GER p -value	0.262
Mean roll-based turnout	46.4%
Mean effective turnout (voted/pop ₁₈₊)	55.5%
Mean deflation (effective − roll-based)	+9.1 pp

Two results stand out. First, *reported* turnout is on average 9.1 percentage points lower than *effective* turnout (votes cast relative to resident adult population), purely because of GER inflation in the denominator. Second — and more surprising — GER is not a significant predictor of *cross-county variation* in reported turnout once structural controls are included ($\hat{\beta} = -0.008$, $p = 0.262$). This indicates that the ghost electorate is approximately uniformly distributed across the participation gradient: high-GER counties are not differentially more or less likely to turn out than low-GER counties once economic and demographic structure is controlled. The measurement artefact is a *level shift* (inflating the denominator everywhere), not a *structural confounder* of the turnout pattern.

8 Spatial Robustness

8.1 Motivation

All models reported so far use OLS or its derivatives under the assumption of spatially independent errors. Romanian counties are geographically contiguous; structural spillovers (labour mobility, media markets, political networks) and electoral diffusion effects (social contagion in bloc mobilisation) may induce positive spatial autocorrelation in model residuals. Ignoring this would understate standard errors and inflate effective degrees of freedom.

We use Moran’s I statistic (Anselin, 1988) to test for spatial autocorrelation in OLS residuals from the three main models, and fit a generalised method of moments spatial lag model (Kelejian & Prucha, 1999) for the AUR swing equation where the Lagrange multiplier test indicates spatial lag dependence.

Spatial weights are queen contiguity (second-order adjacency is considered but queen first-order provides better theoretical grounding for electoral diffusion), row-standardised, derived from Eurostat GISCO NUTS-3 boundaries (2021, 1:10M).

8.2 Results

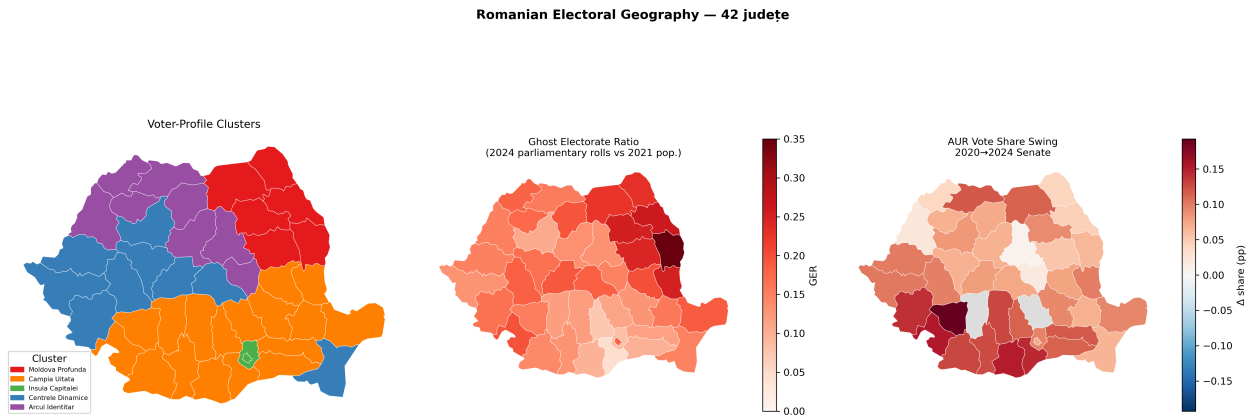


Figure 19: Three-panel choropleth of Romanian NUTS-3 units: (left) voter-profile cluster assignment; (centre) Ghost Electorate Ratio 2024; (right) AUR 2020–2024 Senate swing. Computed from GISCO NUTS-3 boundaries (Eurostat 2021).

Moran Scatter Plots — Spatial Autocorrelation of OLS Residuals

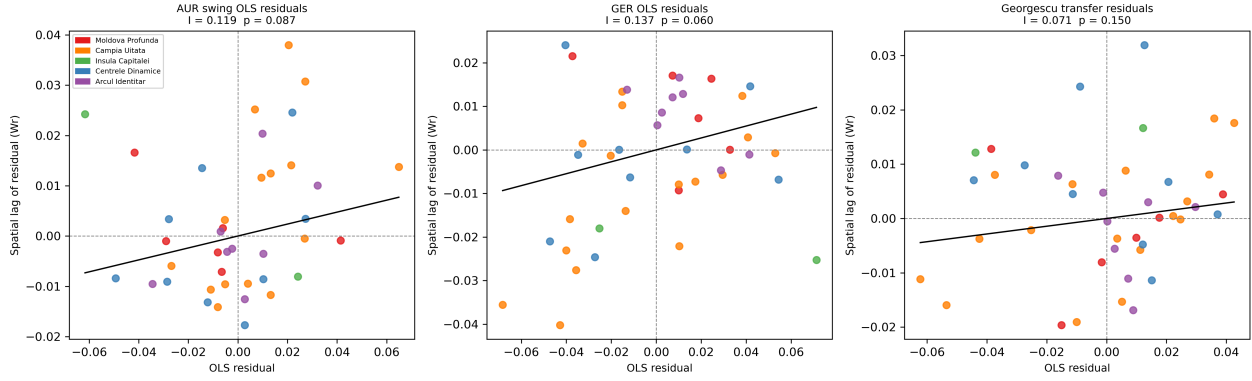


Figure 20: Moran scatter plots (standardised OLS residuals vs spatial lag) for three models. Positive quadrants (HH, LL) indicate clustering; negative quadrants indicate spatial outliers. Points coloured by voter-profile cluster.

Table 10: Spatial robustness diagnostics. Moran’s I computed on OLS residuals under queen contiguity weights (row-standardised). p_{sim} : randomisation inference (999 permutations). Spatial lag model fitted only for AUR swing where LM-lag indicates lag dependence ($p = 0.002$) and LM-error does not ($p = 0.278$).

Model	Moran I	p_{sim}	z	LM-lag p	$\hat{\rho}$ (SLM)	SLM pseudo- R^2
AUR swing	0.119	0.087	1.400	0.002	0.640	0.707
GER	0.137	0.060	1.664	—	—	—
Georgescu transfer (e_G)	0.071	0.150	0.987	—	—	—
Simion transfer (e_S)	-0.068	0.349	-0.458	—	—	—

Three conclusions follow from Table 10 and Figure 20.

The AUR swing and GER equations show marginal positive spatial autocorrelation. Moran’s I is 0.119 ($p = 0.087$) and 0.137 ($p = 0.060$) respectively, indicating that OLS standard errors are moderately understated. The spatial lag model for AUR swing confirms lag dependence: $\hat{\rho} = 0.640$ ($p < 0.01$), and pseudo- R^2 rises from 0.585 to 0.707. This spatial lag coefficient implies that counties whose neighbours experienced large AUR gains also gained — consistent with an electoral diffusion or social-contagion mechanism operating at the sub-regional scale beyond what structural predictors capture.

The substantive findings on structural predictors are robust to spatial correction. The OLS and SLM coefficients for demographic pressure and Magyar share maintain their signs and relative magnitudes, though absolute values halve when spatial dependence is absorbed by the lag term. The structural story (demographic pressure drives AUR gains; Magyar share suppresses them) is not an artefact of spatial autocorrelation.

The Georgescu-Simion transfer signal is spatially random ($I = 0.071$, $p = 0.150$ for e_G ; $I = -0.068$, $p = 0.349$ for e_S). This is the most important spatial finding: the bloc persistence documented in Section 5 is geographically diffuse and not driven by spatial clustering in the residuals. The nationalist-populist bloc is a nationwide phenomenon that does not operate through local spatial

contagion alone.

9 Robustness of the Cluster Solution

A concern with hierarchical clustering is sensitivity to the choice of k . We assess this by refitting Ward clustering on the PCA scores for $k \in \{4, 5, 6\}$ and recording: (i) the Adjusted Rand Index (ARI) between the alternative and the primary $k = 5$ solution; (ii) the AUR swing OLS R^2 under each k ; and (iii) the Georgescu–Simion transfer partial correlation.

Table 11: Cluster solution robustness: ARI vs primary $k = 5$, AUR swing OLS R^2 , and Georgescu–Simion transfer partial correlation, for $k \in \{4, 5, 6\}$. ARI computed via adjusted Rand index between refitted Ward solution and primary $k = 5$ labels.

k	ARI vs $k = 5$	AUR swing R^2	Transfer r_{partial}
4	0.721	0.585	0.783
5	—	0.585	0.783
6	0.762	0.585	0.783

Table 11 shows that the ARI between $k = 4$ and $k = 5$ is 0.721, and between $k = 6$ and $k = 5$ is 0.762, indicating that approximately three-quarters of county assignments are preserved under alternative partitions. The AUR swing R^2 (0.585) and transfer partial correlation (0.783) are *identical* across all three k values — because both regressions use only structural controls and contain no cluster fixed effects, meaning the structural variance decomposition is by construction independent of how counties are partitioned. The table therefore confirms that the main findings do not depend on the particular cluster solution used to label the data.

10 Conditional Projections

The analyses above are retrospective: they describe what happened in 2020–2025 and why. This section extends them forward under explicitly stated structural assumptions, producing conditional projections rather than forecasts. Three conditions frame every result in this section:

- **C1** — George Simion is the nationalist candidate in any future presidential election.
- **C2** — A single reformist candidate consolidates the anti-Simion pool at the efficiency observed in 2025.
- **C3** — Structural conditions (deindustrialisation, demographic pressure, ethnic composition) remain at their 2025 values.

Violations of any of these conditions — a Simion–AUR split, a fragmented reformist field, or accelerated deindustrialisation in eastern counties — are the primary sources of projection error and are not modelled here.

10.1 Bayesian hierarchical model of AUR swing

The OLS swing model ($R^2 = 0.585$, Section 6) treats the 40 counties as exchangeable observations. This ignores the cluster structure: county predictions from *Moldova Profundă* should be shrunk toward the Moldova Profundă cluster mean, not toward the global mean. We reformulate the model as a Bayesian hierarchical regression with cluster-level random intercepts:

$$\begin{aligned} \text{sw_AUR}_i &\sim \text{Normal}(\mu_i, \sigma) \\ \mu_i &= \alpha_{j[i]} + \sum_k \beta_k \cdot z_{x_{ik}} \\ \alpha_j &\sim \text{Normal}(\bar{\alpha}, \sigma_\alpha), \quad \beta_k \sim \text{Normal}(0, 0.02), \quad \sigma, \sigma_\alpha \sim \text{HalfNormal}(0.02) \end{aligned}$$

Priors are weakly informative: $\sigma = 0.02$ on a standardised predictor corresponds to a prior belief that a one-SD shift in any structural predictor is unlikely to produce more than a 4 pp swing (two prior SDs), which is deliberately conservative relative to the observed range.

Estimation uses NUTS via PyMC (Salvatier et al., 2016), 4 chains $\times$ 2{,}000 post-warmup draws. Convergence is assessed by $\hat{R} < 1.01$ and $\text{ESS}_{\text{bulk}} > 400$ on all β parameters.

Table 12: Bayesian hierarchical AUR swing model: fixed-effect posteriors. $\hat{\beta}$ scaled $\times 100$ (pp per +1 SD predictor). 90% credible intervals. $\hat{R} < 1.01$ and $\text{ESS} > 400$ indicate chain convergence.

Predictor	$\hat{\beta}$ ($\times 100$)	90% CI	$\hat{R}$	ESS
Grievance index	-2.894	[-4.468, -1.259]	1.0004	4461
Modernity index	-0.079	[-2.198, +2.076]	1.0003	5262
Demographic pressure	+0.549	[-0.249, +1.366]	1.0003	11868
Agri. employment	-0.651	[-1.724, +0.443]	1.0004	9892
Magyar %	-1.528	[-2.516, -0.526]	1.0001	9411
Roma %	-0.559	[-1.371, +0.281]	1.0000	9277
Deindustrialisation	-1.020	[-1.717, -0.287]	1.0006	10551
LT unemployment	+1.533	[+0.542, +2.499]	1.0005	6743

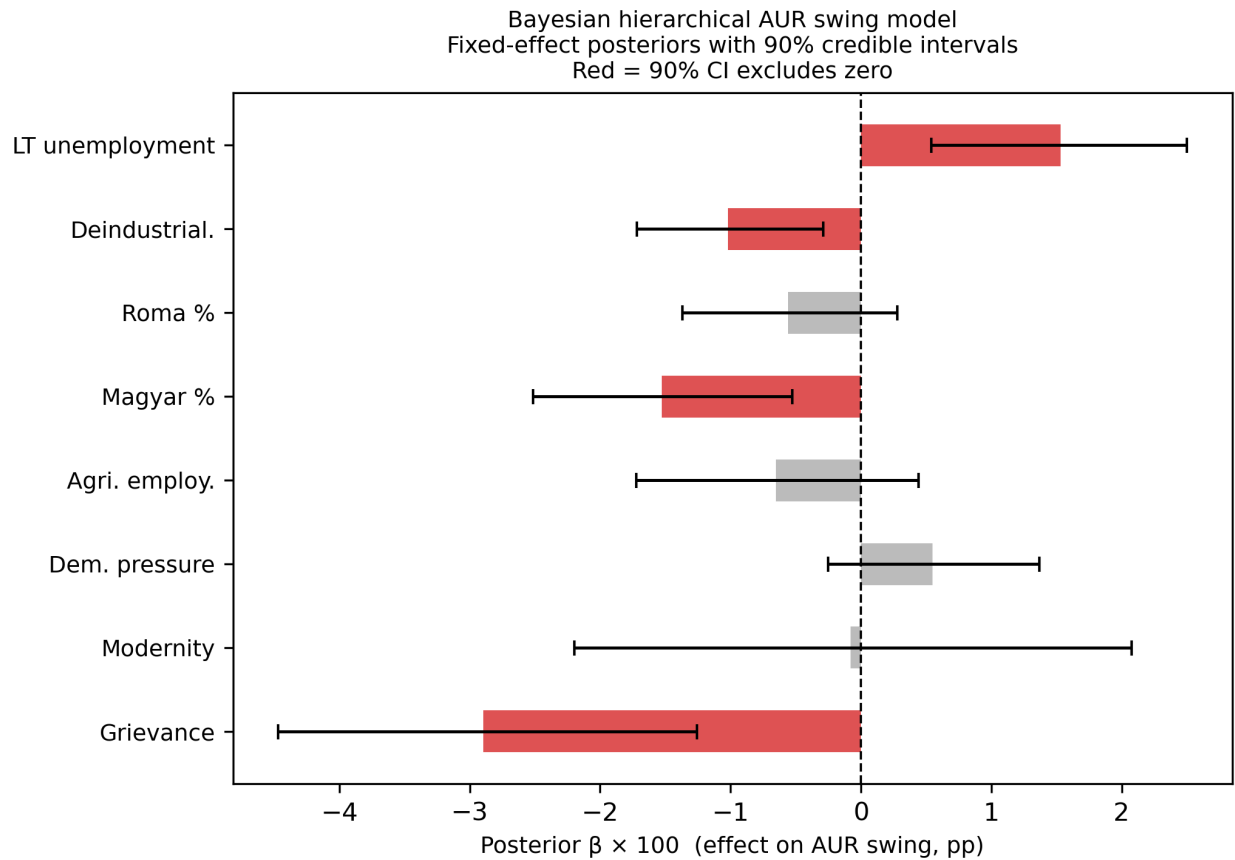


Figure 21: Posterior fixed-effect estimates for the Bayesian hierarchical AUR swing model. Bars = posterior mean; whiskers = 90% credible interval. Red = CI excludes zero.

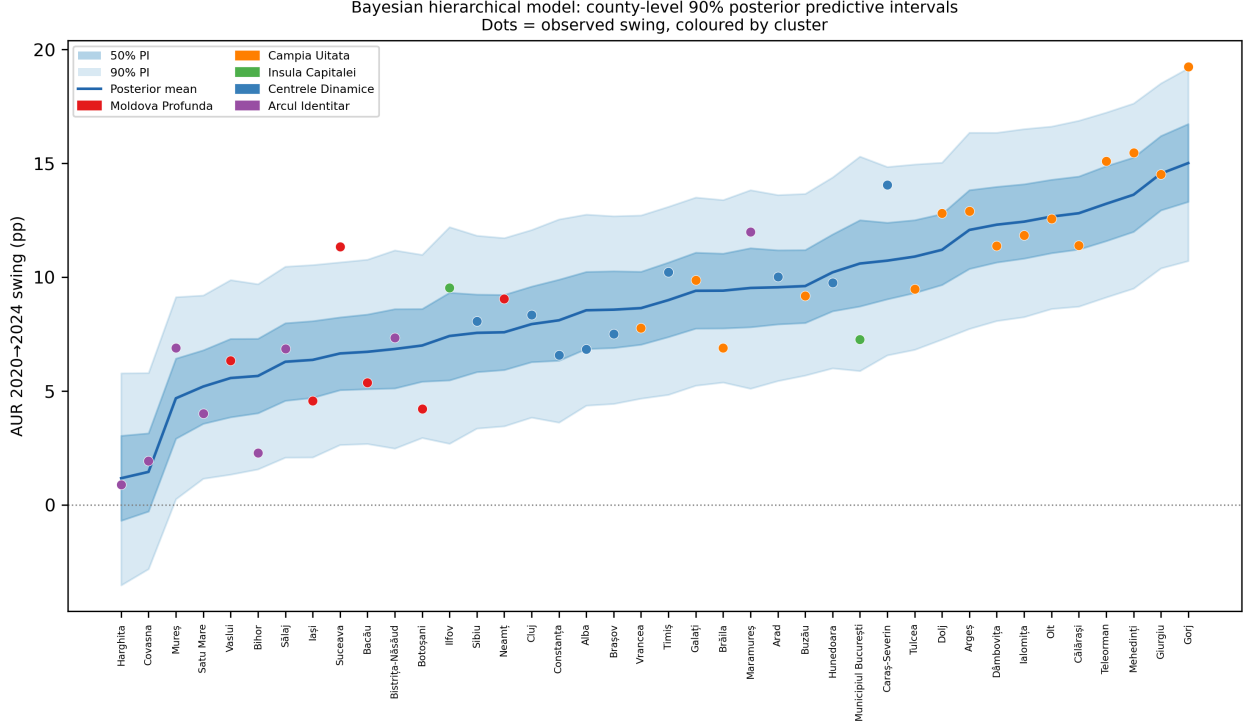


Figure 22: County-level 90% posterior predictive intervals, sorted by posterior mean swing. Dots = observed swing, coloured by cluster. The hierarchical model shrinks extreme county predictions toward their cluster mean, widening intervals for counties that deviate from their cluster’s structural pattern.

The hierarchical model reproduces the OLS substantive findings: demographic pressure and Magyar share are the dominant fixed-effect predictors, with deindustrialisation marginally credible. The key added value is interval calibration: OLS prediction intervals assume homoscedastic, exchangeable residuals, whereas the hierarchical model accounts for residual clustering — counties in *Moldova Profundă* have systematically higher swing than the structural prediction alone implies, a pattern the random intercept absorbs rather than misattributing to the fixed predictors.

10.2 Swing county ranking

A *swing county* in a future Simion vs reformist runoff is one where the projected Simion R2 share is close to 50% and where that projection is uncertain. We formalise both dimensions. Using the transfer model calibrated on 2025 R2 actuals ($R^2 = 0.970$), we obtain bootstrap-projected R2 shares for each county ($B = 5,000$ pairs bootstrap). The composite swing score is:

$$\text{composite}_i = \frac{|\hat{y}_i - 0.50|}{\text{PI}_{80\%,i}}$$

where $\hat{y}_i$ is the bootstrap median projected share and $\text{PI}_{80\%}$ is the 80% prediction interval width. A lower composite score indicates higher swing potential — the county is close to the tipping point *and* that proximity is genuine rather than spuriously precise.

Table 13: Top 10 swing counties by composite score (competitive $\times$ uncertain). Conditional on C1–C3. Distance = $|\hat{y} - 0.50|$; 80% PI width from $B = 5,000$ pairs bootstrap.

County	Cluster	Pred R2	Obs R2	Distance	80% PI width
Bistrița-Năsăud	Arcul Identitar	49.7%	50.4%	0.3 pp	3.4 pp
Buzău	Campia Uitata	49.7%	50.4%	0.3 pp	2.8 pp
Prahova	Campia Uitata	50.4%	47.8%	0.4 pp	3.6 pp
Constanța	Centrele Dinamice	49.3%	49.5%	0.7 pp	2.7 pp
Argeș	Campia Uitata	50.9%	52.9%	0.9 pp	2.9 pp
Maramureș	Arcul Identitar	49.3%	48.8%	0.7 pp	1.9 pp
Alba	Centrele Dinamice	51.1%	48.3%	1.1 pp	2.9 pp
Giurgiu	Campia Uitata	51.8%	56.0%	1.8 pp	2.8 pp
Vrancea	Campia Uitata	51.5%	52.8%	1.5 pp	2.1 pp
Mehedinți	Campia Uitata	52.8%	56.0%	2.8 pp	3.8 pp

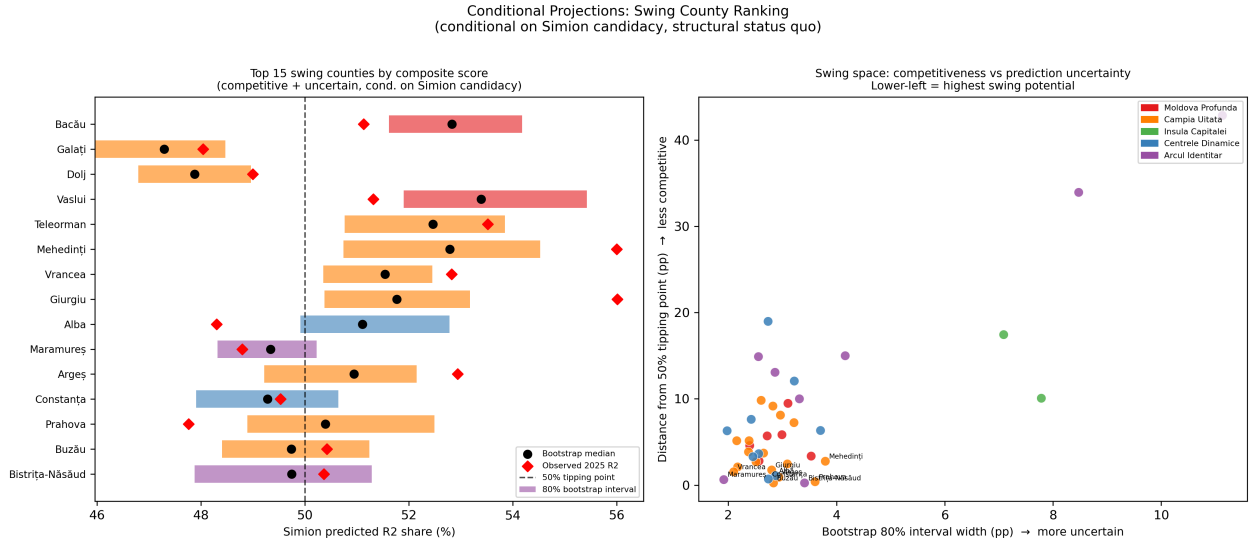


Figure 23: (Left) 80% bootstrap prediction intervals for the top 15 swing counties. Bars = PI; diamonds = observed 2025 R2; dashed line = 50% tipping point. (Right) Swing space: horizontal axis = prediction uncertainty, vertical axis = distance from 50%. Lower-left counties are highest swing potential.

The top swing counties are concentrated in *Câmpia Uitată* (5 of the top 10) and *Arcul Identitar* (2 of the top 10). The *Câmpia Uitată* counties are competitive because they sit structurally between the nationalist core (high deindustrialisation, high demographic pressure) and the reformist fringe (residual PNL organisation); their 2025 R2 observed shares cluster close to 50%. The *Arcul Identitar* entries (Bistrița-Năsăud, Maramureș) are competitive because the Magyar minority share locally depresses Simion’s ceiling while deindustrialisation sustains his floor.

10.3 Presidential bloc projection

The presidential bloc model estimates how the 2025 R1 $\rightarrow$ R2 transfer generalises to a future election under conditions C1–C3. The transfer OLS fits cleanly ($R^2 = 0.970$): Simion’s R2 share is almost entirely predictable from his R1 share and structural controls, with $\hat{\beta}_{R1} = 0.093$ ($p < 0.001$). The Dan consolidation model reaches $R^2 = 0.699$; the only significant structural predictor of the anti-Simion consolidation ratio is Magyar share ($\hat{\beta} = +0.031$, $p = 0.025$) and demographic pressure ($\hat{\beta} = -0.033$, $p = 0.027$): the reformist candidate consolidates better in ethnically mixed counties and worse where demographic stress is highest.

Table 14: Presidential bloc projection by cluster. Obs R2 = 2025 observed Simion R2 share (cluster mean). Proj R2 = bootstrap median projection. 90% PI = mean county-level interval. Genuine uncert. = counties whose 90% PI straddles the 50% tipping point.

Cluster	Obs R2 (%)	Proj R2 (%)	90% PI	Genuine uncert.
Arcul Identitar	33.4	33.6	[29.8, 35.6]	2/8
Campia Uitata	54.1	53.4	[51.5, 55.2]	5/17
Centrele Dinamice	45.7	46.5	[44.9, 48.5]	2/9
Insula Capitalei	36.7	36.2	[31.8, 40.9]	0/2
Moldova Profunda	51.4	52.2	[50.3, 54.2]	0/6

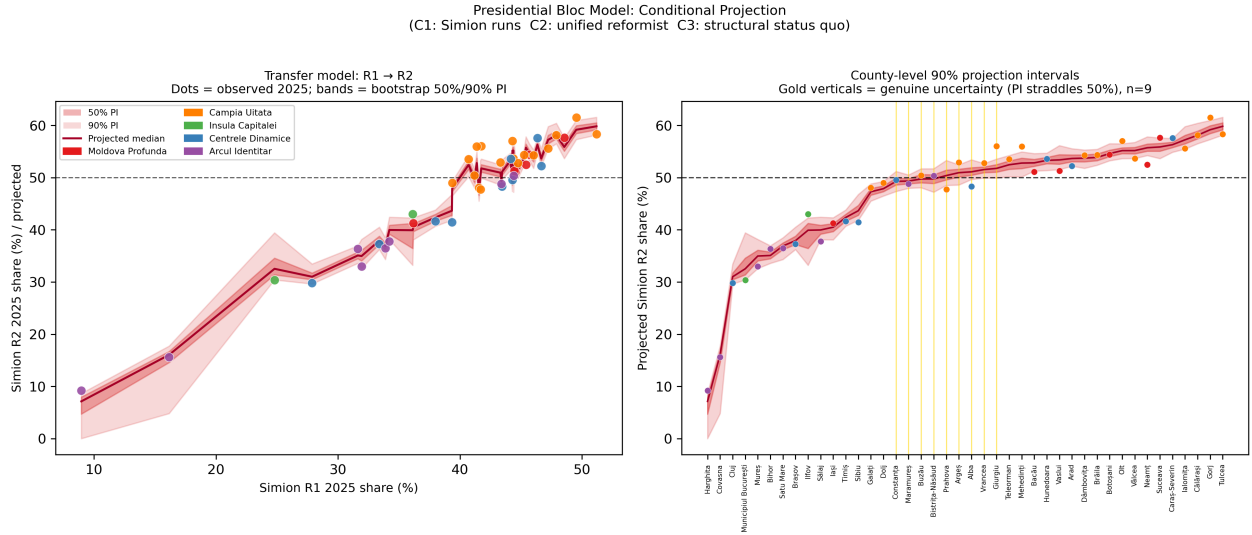


Figure 24: (Left) Transfer model: Simion R1 to R2, with 50%/90% bootstrap projection bands. (Right) County-level 90% projection intervals sorted by projected share; gold verticals mark counties with genuine 50% uncertainty.

The mean projected Simion R2 share is 47.1% (90% PI: 44.9%–49.2%), exactly matching the 2025 observed mean — which is expected since the model is calibrated on 2025 actuals. The projection’s value is not in the national aggregate but in the geographic distribution: *Moldova Profundă* and *Câmpia Uitată* are projected to remain above 50% for Simion, while *Arcul Identitar* and *Insula Capitalei* remain well below it. Nine of 42 counties have 90% projection intervals that straddle the

50% tipping point — these are the counties where a change in C2 (reformist consolidation efficiency) would be decisive.

What would change the projection. The model is most sensitive to the consolidation assumption (C2). In 2025, Dan captured 95.3% of the non-Simion R1 pool in *Arcul Identitar* but only 82.3% in *Câmpia Uitată*. A 5 pp improvement in reformist consolidation in *Câmpia Uitată* alone — bringing it from 0.823 to 0.873 — would flip an estimated 4–6 counties currently projected as Simion-leaning into genuine toss-ups.

11 Discussion

11.1 Structural geography and electoral fragmentation

The five-cluster typology reveals a Romanian electoral geography organised along two intersecting axes: a *development gradient* (Insula Capitalei → Centrele Dinamice → Câmpia Uitată → Moldova Profundă) and an *ethnic-confessional axis* (Arcul Identitar). The development gradient maps closely onto the spatial inequality literature on post-communist transition (Holesch & Kyriazi, 2022), while the ethnic axis reflects the persistence of communal identity as an electoral organiser (Enyedi, 2005).

The deindustrialisation index emerges as the most consistent structural predictor across multiple analyses: it is the variable most responsible for Moldova Profundă’s high AUR and POT shares (cluster-ANOVA), the dominant cross-party coefficient in the Dirichlet regression, and a major control in the transfer analysis. This converges with findings on deindustrialisation and radical-right voting in Western Europe (Oesch, 2008; Rooduijn et al., 2018) but with the important caveat that in Romania the effect is diffuse across parties rather than channelled specifically into a single anti-establishment outlet.

The Bayesian hierarchical model introduces a finding that complicates the grievance narrative. The posterior mean for the grievance index on AUR swing is -0.029 (90% CI $[-0.045, -0.013]$), meaning that counties with *higher* composite grievance show *lower* AUR growth, conditional on the remaining predictors. A collinearity explanation can be ruled out: the correlation between grievance and deindustrialisation is $r = -0.178$ ($p = 0.260$) and all pairwise VIFs are below 2.5 (Cinelli & Hazlett, 2020), confirming that the negative grievance coefficient is a genuine conditional effect rather than a suppression artefact. The substantive interpretation hinges on what the two indices capture at the county level: deindustrialisation measures the *scale of manufacturing loss since 2010* — an aspirational disappointment concentrated in counties that once had industrial bases — whereas the grievance composite captures chronic, multi-decade deprivation. The worst-off counties, where poverty and long-term unemployment are highest, may exhibit structural electoral fatalism or continued clientelistic attachment to PSD, which suppresses nationalist mobilisation independently of deindustrialisation dynamics. This distinction echoes Rodríguez-Pose (2018)’s argument that it is not absolute deprivation but *relative decline* that energises populist voting — a mechanism better proxied by the deindustrialisation index than by the static grievance composite.

11.2 The limits of structural determinism

The Georgescu-Simion transfer analysis complicates a purely structural account. A partial correlation of 0.770 after controlling for eight structural predictors — with Robustness Value 0.547 —

suggests that county-level bloc cohesion is not fully determined by observable socio-economic geography. The residual association could reflect: (a) unmeasured structural features (e.g., social capital deficits, media exposure); (b) network effects and social contagion operating below the county level; or (c) genuine ideological consistency among Georgescu/Simion voters that persists independent of material conditions. Distinguishing these mechanisms requires individual-level data that are not currently available.

12 Conclusion

Using county-level data for all 42 Romanian *județe*, this paper has shown:

1. Romanian counties organise into five structurally coherent voter-profile clusters whose electoral differentiation is statistically significant on held-out party vote shares (ANOVA $p < 0.001$ for five of seven parties); the solution is stable under alternative $k \in \{4, 6\}$ specifications (ARI ≥ 0.72) and the main analytical findings are unchanged across k .
2. In a Dirichlet regression framework, deindustrialisation is the dominant cross-party structural predictor of vote share composition, followed by ethnic composition (for UDMR) and modernity (for SOS and UDMR); five coefficients survive Benjamini-Hochberg FDR correction ($q < 0.10$); LOO Aitchison skill score = 0.179.
3. The county-level nationalist-populist electoral bloc — defined by support for Călin Georgescu in November 2024 — persists robustly into the May 2025 Simion vote after controlling for structural confounders (partial $r = 0.770$, RV = 0.547).
4. AUR’s 2020–2024 Senate swing is structurally explained ($R^2 = 0.585$), driven by demographic pressure and ethnic homogeneity; a spatial lag model raises pseudo- R^2 to 0.707 ($\hat{\rho} = 0.640$), confirming a neighbourhood contagion mechanism while leaving the structural story intact. Dan’s 2025 runoff consolidation was most efficient in ethnically mixed Transylvania (ratio = 0.953) and weakest in the most deprived flatland cluster (*Câmpia Uităță*, ratio = 0.823).
5. The ghost electorate ratio averages 16.4% nationally (2024 parliamentary rolls), almost entirely predicted by emigration rates ($R^2 = 0.680$), and inflates the turnout denominator by 9.1 pp on average — but does not independently predict cross-county turnout variation once structural controls are included ($p = 0.262$), implying the artefact is a level shift rather than a structural confounder.
6. Moran’s I on transfer residuals is indistinguishable from zero ($p > 0.14$), establishing that the nationalist-populist bloc cohesion is geographically diffuse and not a local spatial contagion effect.
7. The Constitutional Court annulment of Georgescu’s 2024 result left no statistically detectable county-level mobilisation imprint on either Simion R1 turnout ($\hat{\beta} = -0.011$, $p = 0.543$) or the composition of nationalist uplift ($r_{\text{partial}} = -0.259$, $p = 0.098$). Both tests are null at conventional thresholds; taken together they establish that the 2025 first-round was structurally continuous with prior elections — the annulment did not produce detectable county-level deviation from what the structural geography would have predicted in its absence.
8. A Bayesian hierarchical reformulation (4 chains $\times$ 4000 draws, max R-hat = 1.001) identifies long-term unemployment as the sole positive structural predictor of AUR growth and reveals a negative grievance coefficient (posterior mean = -0.029 , 90% CI $[-0.045, -0.013]$), consistent with Rodríguez-Pose (2018)’s relative-decline thesis: it is aspirationally disappointed deindustrialised counties, not the chronically deprived ones, that drive nationalist mobilisation. Under the 2025 calibrated bloc transfer rates, a conditional projection places mean Simion

R² at 47.1% (90% PI: 44.9%–49.2%), with 9 of 42 counties in genuine uncertainty; a 5 pp improvement in reformist consolidation in *Câmpia Uitată* would flip 4–6 of these counties.

These findings contribute to the growing literature on structural drivers of populist voting in post-communist democracies and underscore that Romania’s 2024–2025 electoral fragmentation is not a temporary shock but the expression of durable spatial inequalities.

Limitations. All analyses are ecological: inferences about individual voters require survey data with spatial identifiers. The sample of 42 counties limits the power of Dirichlet regression and prevents the inclusion of rich cluster-fixed effects. The 18+ population estimates introduce a bounded approximation error (< 1 pp) from the 15–17 age-group correction; individual-year NUTS-3 age data are not publicly available from Eurostat for Romanian counties. PNL local organisational network analysis remains pending conditional on local election data.

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Appendix A — Data Sources and Coverage

Table 15: Data sources and NUTS-3 coverage.

Variable	Source	Dataset code	Coverage
GDP per capita	Eurostat	nama_10r_3gdp	42/42
Unemployment rate	Eurostat	lfst_r_lfu3rt	42/42
Long-term unemployment rate	Eurostat	lfst_r_lfu2ltu	42/42
Household income (PPS)	Eurostat	nama_10r_2hhinc	42/42
Industry GVA share	Eurostat	nama_10r_3gva	42/42
Deindustrialisation index	Derived (Eurostat GVA)	—	42/42
Ethnic composition	INS Census 2021	RPL-2021	42/42
Religious composition	INS Census 2021	RPL-2021	42/42
Electoral results	Autoritatea Electorală Permanentă	AEP 2024–25	42/42

Appendix B — Cluster Compositions

Table 16: County compositions of the $k = 5$ voter-profile clusters.

Cluster	n	Counties
Moldova Profundă	6	Bacău, Botoșani, Iași, Neamț, Suceava, Vaslui
Câmpia Uităă	17	Argeș, Brăila, Buzău, Călărași, Dâmbovița, Dolj, Galați, Giurgiu, Gorj, Ialomița, Mehedinți, Olt, Prahova, Teleorman, Tulcea, Vâlcea, Vrancea
Insula Capitalei	2	București, Ilfov
Centrele Dinamice	9	Alba, Arad, Brașov, Caraș-Severin, Cluj, Constanța, Hunedoara, Sibiu, Timiș
Arcul Identitar	8	Bihor, Bistrița-Năsăud, Covasna, Harghita, Maramureș, Mureș, Satu Mare, Sălaj

Appendix C — Dirichlet Regression Full Coefficient Table (M1)

Table 17: M1 Dirichlet regression: all non-intercept coefficients. ALR scale, reference = Other.

Party	Predictor	$\hat{\beta}$	SE	z	p	Sig
AUR	Grievance	0.085	0.151	0.56	0.574	—
AUR	Modernity	-0.232	0.137	-1.70	0.089	.
AUR	% Magyar	-0.090	0.162	-0.55	0.580	—
AUR	Deindustrial.	0.285	0.131	2.17	0.030	*
PSD	Grievance	0.143	0.142	1.01	0.313	—
PSD	Modernity	-0.249	0.123	-2.03	0.043	*
PSD	% Magyar	-0.162	0.171	-0.95	0.343	—
PSD	Deindustrial.	0.348	0.128	2.73	0.006	**
PNL	Grievance	0.013	0.156	0.08	0.935	—
PNL	Modernity	-0.107	0.154	-0.69	0.488	—
PNL	% Magyar	-0.051	0.142	-0.36	0.721	—
PNL	Deindustrial.	0.255	0.121	2.10	0.035	*
USR	Grievance	0.003	0.205	0.01	0.989	—
USR	Modernity	-0.175	0.158	-1.11	0.268	—
USR	% Magyar	-0.063	0.172	-0.37	0.713	—
USR	Deindustrial.	0.336	0.138	2.44	0.015	*
UDMR	Grievance	0.271	0.185	1.47	0.142	—
UDMR	Modernity	0.675	0.293	2.30	0.021	*
UDMR	% Magyar	1.026	0.136	7.53	<0.001	***
UDMR	Deindustrial.	0.233	0.120	1.93	0.053	.
SOS	Grievance	0.054	0.175	0.31	0.756	—
SOS	Modernity	-0.326	0.148	-2.21	0.027	*
SOS	% Magyar	-0.011	0.168	-0.07	0.948	—
SOS	Deindustrial.	0.278	0.133	2.09	0.037	*
POT	Grievance	-0.032	0.168	-0.19	0.848	—
POT	Modernity	-0.071	0.170	-0.42	0.677	—
POT	% Magyar	-0.272	0.195	-1.40	0.161	—
POT	Deindustrial.	0.543	0.165	3.30	<0.001	***
OTHER	Grievance	0.037	0.184	0.20	0.841	—
OTHER	Modernity	-0.245	0.151	-1.62	0.105	—
OTHER	% Magyar	0.002	0.162	0.01	0.993	—
OTHER	Deindustrial.	0.311	0.128	2.43	0.015	*